Introduction to superconductivity
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Borja Álvarez Reguera

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The evolution of superconductivity is characterized by profound discoveries and persistent scientific challenges. Initiated by the pioneering 1911 experiments of Kamerlingh Onnes regarding the electrical resistivity of mercury, the field underwent a transformative expansion following the 1986 discovery of high-T_c superconductivity in Ba-doped lanthanum cuprate by Bednorz and Müller. At its core, **the phenomenon arises from a subtle effective attractive interaction between charge carriers**—either conduction electrons or valence holes—leading to the formation of pairs at cryogenic temperatures. This pairing mechanism governs the fundamental signatures of the superconducting state, including perfect conductivity, the Meissner effect (perfect diamagnetism), anomalous thermodynamic behavior, and flux quantization. While initial progress relied on empirical observations, the seminal Bardeen-Cooper-Schrieffer (BCS) theory of 1957 successfully integrated these diverse phenomena into a rigorous theoretical framework.

This work prioritizes the most significant experimental and theoretical pillars of the field, with a particular emphasis on the many-body BCS microscopic theory and the Ginzburg-Landau (GL) phenomenological approach. The GL theory, in particular, offers a powerful framework for analyzing phase transitions and superconducting effects via an order parameter, bypassing the need for explicit many-body wavefunctions while remaining grounded in BCS principles. By bridging phenomenological descriptions—such as the London model and Josephson effects—with microscopic insights, **this text provides a self-contained technical foundation**. For those new to the subject, it is recommended to first explore the descriptive sections on Cooper pairing and macroscopic quantum phenomena to better appreciate the logical coherence of the complete theoretical structure.

1. Phenomenological aspects.

Old and new superconducting materials.

Superconductivity was discovered in 1911 by Kamerlingh Onnes shortly after his successful liquefaction of helium. While investigating the electrical resistance of mercury at cryogenic temperatures, he observed that at approximately 4.2 K, the resistance dropped sharply within a range of 0.01 K to non-measurable values. This resistanceless state is characterized by an actual resistivity $\rho = 0$ for continuous currents significantly lower than a critical value, provided the temperature remains well below T_c and external magnetic fields are absent. A notable consequence is that a current induced in a closed superconducting ring can circulate for years without detectable decay, a phenomenon that also entails the quantization of the magnetic flux threading the ring. It is important to emphasize that dissipationless current flow alone does not uniquely define superconductivity—as seen in the quantum Hall regime—but **rather the combination of perfect conduction and the Meissner effect** (the expulsion of weak magnetic fields from the bulk). In superconducting rings, flux quantization is a direct manifestation of this effect.

Historically, the search for higher transition temperatures was slow; niobium held the elemental record at $T_c = 9.25$ K, and the alloy Nb_3Ge maintained a peak of $T_c = 23.3$ K from 1972 until 1986. The discovery of La-based cuprates by Bednorz and Müller, with $T_c > 30$ K, catalyzed a rapid acceleration in the field. This was soon followed by the discovery of cuprate families with critical temperatures exceeding 77 K, the boiling point of liquid nitrogen, transitioning high-T_c superconductivity from theoretical speculation to experimental reality.

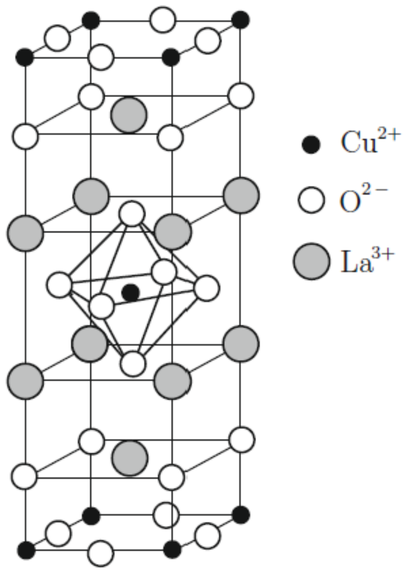


Figure 1: Crystal structure of the ceramic material La_2CuO_4. Appropriately doped, lanthanum-based cuprates opened the path to high-T_c superconductivity in 1986.

The copper oxide superconductors are defined by crystal structures containing one or more CuO_2 sheets separated by insulating block layers. In the parent compound La_2CuO_4, which possesses a body-centered tetragonal structure (see Fig. 1), the CuO_2 planes are spaced ~ 6.6 Å apart and separated by two LaO planes. Each copper ion is coordinated by an elongated oxygen octahedron, with Cu−O distances of ~ 1.9 Å within the plane and ~ 2.4 Å apically.

The electronic configuration of the constituent atoms (La: $5d^1 6s^2$; O: $2s^2 2p^4$; Cu: $3d^{10} 4s^1$) dictates the material's properties. In the crystal lattice, lanthanum adopts a La^{3+} closed-shell configuration, and oxygen becomes O^{2-}. To maintain charge neutrality, copper takes the ionic configuration Cu^{2+}: $3d^9$. Although the resulting hole in the d-shell suggests metallic behavior in an independent-electron model, La_2CuO_4 is a Mott-Hubbard insulator due to strong Coulomb repulsions that localize the Cu^{2+} $3d$ holes. Furthermore, the magnetic moments of these ions are coupled antiferromagnetically via super-exchange through the oxygen atoms, leading to an antiferromagnetic insulating state with a Néel temperature of ~ 320 K.

The transition from an insulating to a superconducting state in $La_{2-x}Sr_xCuO_4$ is achieved through chemical doping, where La^{3+} ions are randomly substituted by Sr^{2+} or Ba^{2+} ions. This substitution reduces the number of electrons donated to the CuO_2 planes, resulting in the generation of mobile holes. These carriers reside primarily in the O $2p$ orbitals, which are hybridized with the Cu $3d_{x^2-y^2}$ orbitals. Physically, $La_{2-x}Sr_xCuO_4$ can be described by a localized model for the Cu^{2+} states and an itinerant model for the oxygen $2p$ holes. The superconducting phase emerges within the doping range $0.05 < x < 0.30$, with an optimal concentration of $x \sim 0.15$ yielding a maximum critical temperature $T_c \sim 38$ K.

The success of lanthanum-based compounds led to the identification of other cuprate families involving yttrium, bismuth, thallium, and mercury. Despite their apparent complexity, the fundamental structure of these cuprates consists of an alternating sequence of electronically active metallic CuO_2 layers and insulating block layers. These block layers serve as "charge reservoirs," supplying either electrons or holes to the active planes. In such lamellar copper oxides, critical temperatures $T_c \sim 100$ K or higher are frequently observed (see Table 1), with Tl-bearing and Hg-bearing compounds representing some of the most promising materials for high-temperature applications.

In 1991, fullerene (C_{60}) was introduced as a novel superconducting material. Solid C_{60} is composed of molecules arranged in an fcc lattice and held together by weak van der Waals forces. Upon doping with alkali metals, such as in K_3C_{60}, excess electrons populate the Lowest Unoccupied Molecular Orbital (LUMO), rendering the solid conductive. K_3C_{60} exhibits a $T_c \sim 18$ K, a remarkably high value for a molecular superconductor compared to other intercalated systems like graphite.

Further advancements include the discovery of superconductivity in MgB_2 with $T_c \sim 40$ K, which highlighted the significance of two-band and multiband electronic structures. Additionally, the emergence of iron-based layered superconductors, or oxypnictides such as $LaO_{1-x}F_xFeAs$, has reached $T_c \sim 50$ K. These materials, along with heavy fermions and organic crystals, continue to broaden the theoretical and applicative scope of the field.

Material	T_c **(K)**	**Critical Field (0)**
Metallic elements		H_c *(Gauss)*
Al	1.17	105
Sn	3.72	305
Pb	7.19	803
Hg	4.15	411
Nb	9.25	2060
V	5.40	1410
Binary compounds		H_{c2} *(Tesla)*
V_3Ga	16.5	27
V_3Si	17.1	25
Nb_3Al	20.3	34
Nb_3Ge	23.3	38
MgB_2	40	$\sim 5; \sim 20$
Other compounds		H_{c2} *(Tesla)*
UPt_3 (heavy fermion)	0.53	2.1
$PbMo_6S_8$ (Chevrel phase)	12	55
κ-$[BEDT-TTF]_2Cu[NCS]_2$	10.5	~ 10
Rb_2CsC_{60} (fullerene)	31.3	~ 30
$NdFeAsO_{0.7}F_{0.3}$ (pnictide)	47	$\sim 30; \sim 50$
Cuprate oxides		H_{c2} *(Tesla)*
$La_{2-x}Sr_xCuO_4$ ($x \sim 0.15$)	38	~ 45
$YBa_2Cu_3O_7$	92	~ 140
$Bi_2Sr_2CaCu_2O_8$	89	~ 107
$Tl_2Ba_2Ca_2Cu_3O_{10}$	125	~ 75

Table 1: Critical temperature T_c and zero-temperature critical field $H_c(0)$ or $H_{c2}(0)$ for selected superconductors.

General considerations and some phenomenological aspects.

The physical origin of superconductivity lies in an effective attractive interaction between charge carriers—conduction electrons in n-type materials or valence holes in p-type conductors. While the following discussion focuses on electron carriers for brevity, the principles remain analogous for hole-driven systems. In "conventional" superconductors, this subtle attraction near the Fermi level is mediated by the phonon field, whereas in "non-conventional" materials, alternative pairing mechanisms may be present. Regardless of the specific mediator, the Bardeen-Cooper-Schrieffer (BCS) theory provides the unified theoretical framework necessary to interpret these states.

At temperatures exceeding T_c, this attractive interaction typically results only in minor superconducting fluctuations. However, below the critical temperature, it leads to the **formation of highly correlated electron pairs, known as Cooper pairs**, within an energy shell surrounding the Fermi surface. This correlated gas exhibits properties fundamentally distinct from a standard non-interacting electron gas. The pairing mechanism is responsible not only for perfect conductance but also for perfect diamagnetism, anomalous specific heat, transport property modifications, and the emergence of an energy gap in the quasiparticle spectrum.

A defining characteristic of the superconducting state, beyond the absence of resistance, is perfect diamagnetism. As discovered by Meissner and Ochsenfeld in 1933, a magnetic field below a certain critical threshold is entirely expelled from the bulk of a superconductor (see Fig. 2). This Meissner effect is distinct from the behavior of a hypothetical "perfect conductor," which would merely trap the magnetic flux present at the moment of the transition. In a true superconductor, the field is expelled upon cooling below T_c regardless of the initial state. This unique link between perfect conductivity and perfect diamagnetism is a direct consequence of the pairing mechanism and the resulting coherent quantum state.

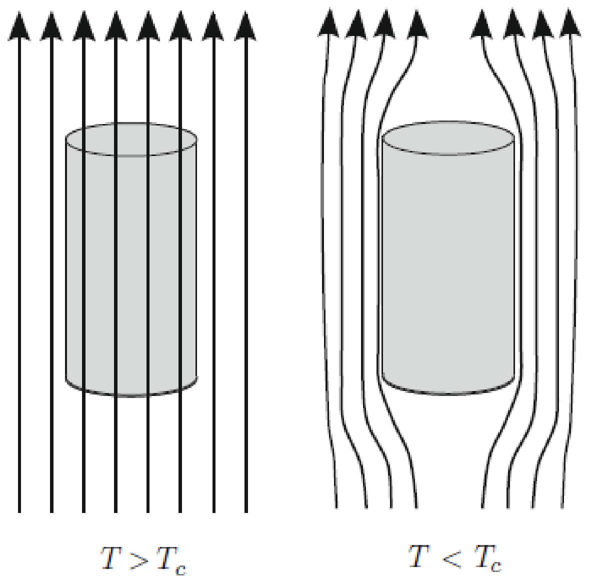


Figure 2: Schematic representation of the Meissner-Ochsenfeld effect. The magnetic field lines are expelled from the interior of the sample as it is cooled below the critical temperature T_c, demonstrating perfect diamagnetism.

The exclusion of a magnetic field from the interior of a superconductor arises from lossless "diamagnetic screening currents" that flow within a thin surface layer approximately 10^{-5} cm thick. Although these self-generated supercurrents are physically responsible for maintaining $B = 0$ in the bulk, it is formally convenient to adopt a diamagnetic description by assigning an internal magnetization M (magnetic moment per unit volume). Given that the magnetic induction is defined by

$$\vec{B} = \vec{H} + 4\pi\vec{M} \tag{1}$$

the vanishing of B inside the superconductor implies a static magnetic susceptibility $\chi = \frac{M}{H} = -\frac{1}{4\pi}$. Comparing this to the orbital diamagnetism of a normal metal ($\chi \sim -10^{-6}$), the superconducting response is approximately six orders of magnitude larger, identifying the material as a "superdiamagnet." The energy per unit volume required for the screening currents to expel the field is given by

$$-\int_0^H M dH = \frac{H^2}{8\pi} \tag{2}$$

This robust diamagnetic behavior enables phenomena such as magnetic levitation, now a standard demonstration using high-T_c materials cooled by liquid nitrogen. While weak fields are entirely expelled, sufficiently high magnetic fields eventually destroy the superconducting state. The nature of the transition to the normal state depends on both intrinsic material properties and sample geometry, the latter characterized by demagnetization factors. To investigate purely microscopic properties, it is conventional to consider long, thin cylindrical rods with their axes parallel to the applied field, a "standard geometry" where demagnetization factors are zero. Based on their response to magnetic fields, superconductors are classified into two categories: type-I (or soft) and type-II (or hard) superconductors.

Type-I superconductors.

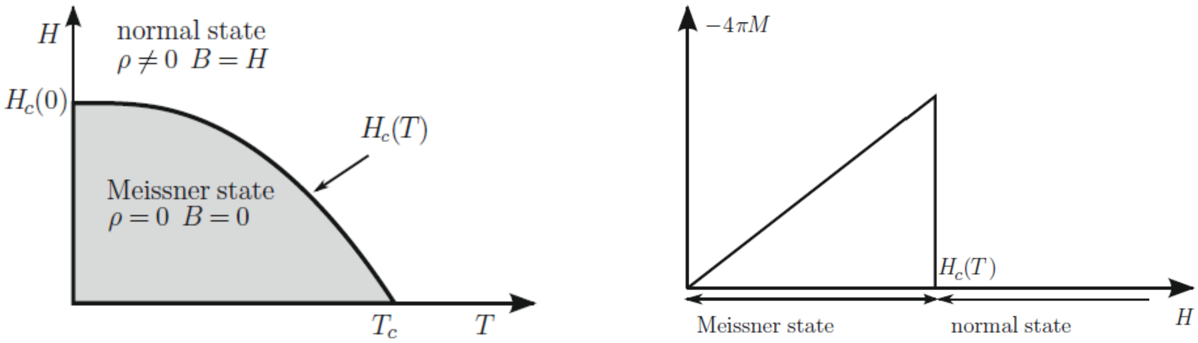
Type-I superconductors are characterized by a **direct transition** between the superconducting and normal states under the influence of an external magnetic field H at a temperature T. When the applied field is lower than a specific critical value $H_c(T)$, the material remains in the Meissner state, completely expelling the magnetic flux from its interior. Once the external field exceeds $H_c(T)$, the entire specimen reverts to the normal state.

The equilibrium curve $H_c(T)$ defines the boundary between the "normal" and "superconducting" regions in the $H - T$ phase diagram (see Fig. 3a). For many type-I superconductors, the temperature dependence of the critical field is well-approximated by the parabolic relation:

$$H_c(T) = H_c(0)\left[1 - \frac{T^2}{T_c^2}\right] \tag{3}$$

where $H_c(0)$ represents the extrapolated critical field at absolute zero. As the temperature approaches T_c from below, the critical field follows an approximately linear behavior, $H_c(T) \propto (T_c - T)$.

The magnetic response of these materials is further illustrated by the magnetization M as a function of the applied field H in Fig. 3b. In the Meissner state ($H < H_c(T)$), the condition $B = H + 4\pi M = 0$ holds, leading to the linear relation $-4\pi M = H$. For $H > H_c(T)$, the material behaves as a normal metal. In this regime, the small magnetic susceptibility of the normal phase can be neglected, resulting in $B = H$. The majority of elemental superconductors belong to this type-I category.



(a) Phase diagram showing the normal and Meissner regions.

(b) Magnetization M vs. applied field H.

Figure 3: Magnetic properties of type-I superconductors: (a) $H - T$ phase diagram and (b) constitutive magnetization curve.

Type-II superconductors.

Type-II superconductors exhibit a more complex magnetic response characterized by two distinct critical fields. At a given temperature T, if the external field H remains below the lower critical field $H_{c1}(T)$, the material stays in the Meissner state, completely expelling the magnetic flux. However, as the field increases within the range $H_{c1}(T) < H < H_{c2}(T)$, **the flux partially penetrates the specimen**, forming a regular triangular lattice of flux-bearing regions. This regime is known as the mixed state, vortex state, or Shubnikov phase. In this phase, the density of these magnetic filaments or vortices increases with the field until the upper critical field $H_{c2}(T)$ is reached, at which point the material transitions entirely to the normal state.

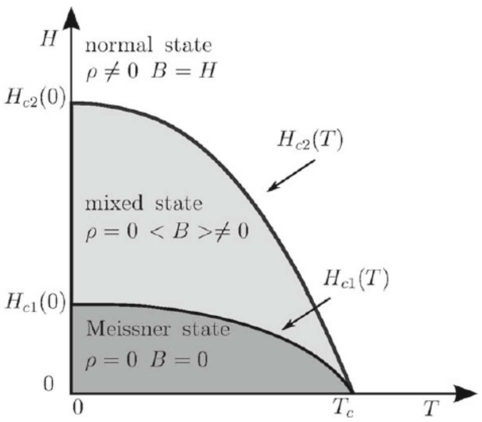
The phase diagram in the $H - T$ plane is thus divided into Meissner, mixed, and normal regions. High-field applications primarily rely on type-II materials because $H_{c2}(T)$ can reach several tens of Tesla. In the mixed state, the material can support high currents with negligible dissipation, provided that material defects—acting as "pinning centers"—prevent the motion of the vortices. Both $H_{c1}(T)$ and $H_{c2}(T)$ generally follow a parabolic temperature dependence similar to the one previously described in Eq. 3.

The magnetization curve $M(H)$ further clarifies these regimes. While perfect diamagnetism persists up to $H_{c1}(T)$, the magnetization gradually decreases in the mixed state until it becomes negligible at $H_{c2}(T)$. This behavior allows for the definition of the thermodynamic critical field $H_c(T)$ through an equivalent area construction, where the low-field linear magnetization is extrapolated such that the total area under the $M(H)$ curve is preserved. Consequently, for type-II superconductors, the relationship $H_{c1}(T) < H_c(T) < H_{c2}(T)$ holds, whereas in type-I materials, these three values coincide.

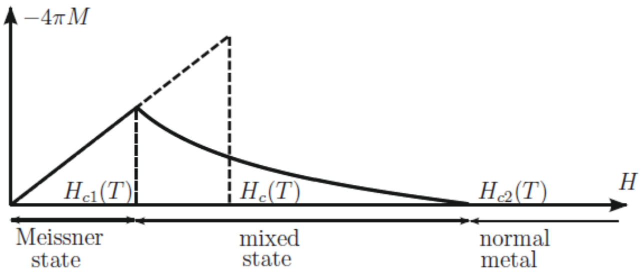
In the mixed state, each vortex consists of a normal metallic core carrying magnetic flux, surrounded by a superconducting region with screening supercurrents. **Each of these filaments carries exactly one quantum of magnetic flux**:

$$\Phi_0 = \frac{hc}{2e} = 2.0679 \times 10^{-7} \text{ gauss} \cdot \text{cm}^2 \tag{18.3}$$

The distinction between type-I and type-II superconductors fundamentally depends on the sign of the surface energy at the superconductor-normal interface, being positive for type-I and negative for type-II. This rich phenomenology sets the stage for a deeper exploration of the microscopic quantum theory.



(a) Phase diagram for type-II superconductors.



(b) Magnetization curve and H_c construction.

Figure 4: Magnetic characteristics of type-II superconductors: (a) $H - T$ boundaries and (b) magnetization behavior showing the mixed state.

2. Cooper pairs.

The existence of an effective attractive interaction between conduction electrons is the fundamental cornerstone of the microscopic interpretation of superconductivity. A primary mechanism for this attraction, as proposed by Fröhlich, is the **indirect electron-electron interaction mediated by the phonon field**. In simple metallic systems, conduction electrons are typically treated as independent fermions weakly coupled to lattice vibrations. Within this framework, an electron polarizes the lattice as it moves, and a second electron subsequently interacts with this polarized state. This second-order process results in a net attractive interaction for electrons located near the Fermi energy E_F within an energy shell of the order of $\hbar\omega_D$, where ω_D denotes the Debye frequency.

While alternative mediators—such as spin fluctuations, bipolarons, or anharmonic vibrational modes—have been suggested to explain high-temperature superconductors, the consequences of this attraction remain universal. A significant advancement in this theory was Cooper's demonstration that the normal Fermi sea is inherently unstable in the presence of even an infinitesimal attractive interaction.

To illustrate this, consider the ground state of a free-electron gas at $T = 0$, where all states are filled up to the Fermi wavevector $\vec{k}_F$. If two "extra" electrons are added to this system, as shown in Fig. 5, they interact via a weak attractive two-body potential $U(\vec{r}_1, \vec{r}_2)$. These electrons are subject to the Pauli exclusion principle, meaning they are restricted from occupying any states already filled within the Fermi sea. The corresponding Schrödinger equation for this two-electron system is given by:

$$\left[\frac{\vec{p}_1^2}{2m} + \frac{\vec{p}_2^2}{2m} + U(\vec{r}_1, \vec{r}_2)\right] \psi(\vec{r}_1\sigma_1, \vec{r}_2\sigma_2) = E\psi(\vec{r}_1\sigma_1, \vec{r}_2\sigma_2) \tag{4}$$

where $\vec{r}_i$ and σ_i represent the spatial and spin coordinates, respectively. Since the Hamiltonian lacks spin-dependent terms, the total wavefunction is separable into spatial and spin parts:

$$\psi(\vec{r}_1\sigma_1, \vec{r}_2\sigma_2) = \phi(\vec{r}_1, \vec{r}_2)\chi(\sigma_1, \sigma_2) \tag{5}$$

In the resulting bound state, the two electrons form what is known as a Cooper pair by scattering between states $(\vec{k}\uparrow, -\vec{k}\downarrow)$ **and** $(\vec{k}'\uparrow, -\vec{k}'\downarrow)$ **through phonon exchange**, effectively lowering the system's energy below $2E_F$.

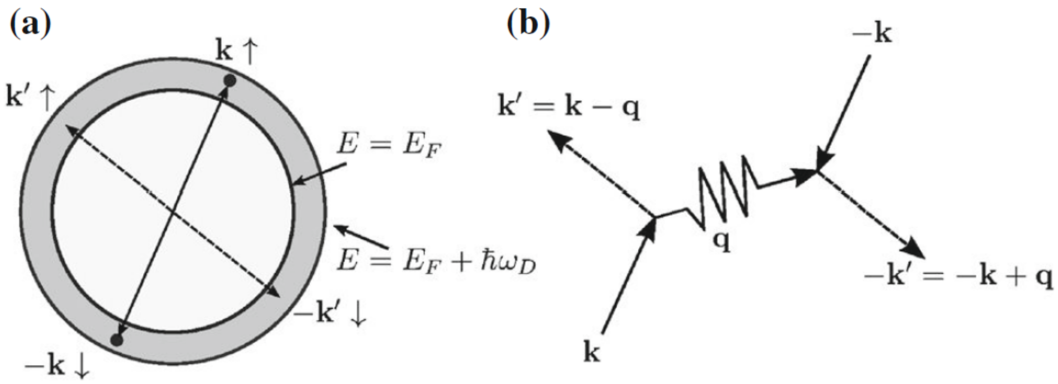


Figure 5: Schematic of the Cooper pair formation. (a) Two electrons with opposite momenta and spins scatter into available states within the energy shell $E_F < E_k < E_F + \hbar\omega_D$. (b) The scattering process mediated by the exchange of a phonon with momentum $\hbar\vec{q}$.

Regarding the spin configuration, we first consider the antisymmetric singlet state, which corresponds to a total spin $S = 0$:

$$\chi^{(S=0)} = \frac{1}{\sqrt{2}}\left(\alpha(1)\beta(2) - \beta(1)\alpha(2)\right) \tag{6}$$

where α and β represent the spin-up and spin-down eigenfunctions, respectively. Conversely, one may consider the symmetric triplet state with total spin $S = 1$. This state comprises three partner functions corresponding to the magnetic projections $M_S = +1, 0, -1$:

$$\chi^{(S=0)} = \begin{cases} \alpha(1)\alpha(2) \\ \frac{1}{\sqrt{2}} \left(\alpha(1)\beta(2) + \beta(1)\alpha(2)\right) \\ \beta(1)\beta(2) \end{cases} \tag{7}$$

According to the Pauli exclusion principle, the total wavefunction $\psi(\vec{r}_1\sigma_1, \vec{r}_2\sigma_2)$ must be antisymmetric. Consequently, the symmetry of the orbital part $\phi(\vec{r}_1, \vec{r}_2)$ is strictly determined by the choice of the spin state in Eq. 6 or 7.

To solve the Schrödinger equation for this system, we expand the orbital component using a basis of plane waves $W(\vec{k}, \vec{r}) = \frac{1}{\sqrt{V}} e^{i\vec{k}\cdot\vec{r}}$. In a **homogeneous medium**, the potential depends only on the relative distance, $U(\vec{r}_1, \vec{r}_2) = U(\vec{r}_1 - \vec{r}_2)$, implying that the total momentum $\hbar\vec{K}$ is a conserved quantity. Seeking the ground state of the pair, we set $\vec{K} = 0$, which dictates that the constituent electrons possess opposite momenta: $\vec{k}_1 = \vec{k}$ and $\vec{k}_2 = -\vec{k}$.

The most general wavefunction for a pair with zero total spin ($S = 0$) and zero total momentum ($\vec{K} = 0$) is expressed as:

$$\psi(\vec{r}_1\sigma_1, \vec{r}_2\sigma_2) = \sum_{\vec{k}} g(\vec{k}) \frac{1}{V} e^{i\vec{k}\cdot(\vec{r}_1 - \vec{r}_2)} \frac{1}{\sqrt{2}} [\alpha(1)\beta(2) - \beta(1)\alpha(2)] \tag{8}$$

In Eq. 8, $g(\vec{k})$ represents the probability amplitude of finding the electrons in the states $\vec{k}$ and $-\vec{k}$. To satisfy the global antisymmetry requirement, $g(\vec{k})$ must be an even function, $g(\vec{k}) = g(-\vec{k})$.

Similarly, for a triplet state ($S = 1$) with a specific spin component (e.g., $S_z = 1$), the wavefunction is:

$$\psi(\vec{r}_1\sigma_1, \vec{r}_2\sigma_2) = \sum_{\vec{k}} g(\vec{k}) \frac{1}{V} e^{i\vec{k}\cdot(\vec{r}_1 - \vec{r}_2)} \alpha(1)\alpha(2) \tag{9}$$

In this case, the spatial antisymmetry requires $g(\vec{k})$ to be an odd function, such that $g(\vec{k}) = -g(-\vec{k})$.

To account for the Pauli exclusion principle between the additional electron pair and the occupied states of the Fermi sea, we impose the condition:

$$g(\vec{k}) = 0 \quad \text{for} \quad E_k = \frac{\hbar^2 k^2}{2m} < E_F \tag{10}$$

where E_k denotes the unperturbed free-electron energy. Furthermore, **following the theory of phonon-mediated coupling**, we assume the potential $U(\vec{r}_1 - \vec{r}_2)$ is attractive only within a narrow energy shell of width $\hbar\omega_D$ around the Fermi level. Consequently, we require:

$$g(\vec{k}) = 0 \quad \text{for} \quad E_k > E_F + \hbar\omega_D \tag{11}$$

In typical metals, the Debye energy satisfies the condition $\hbar\omega_D \ll E_F$. The matrix elements of the potential between an initial state $(\vec{k}', -\vec{k}')$ and a final state $(\vec{k}, -\vec{k})$ are defined as:

$$U_{\vec{k}\vec{k}'} = \iint \frac{1}{V} e^{-i\vec{k}\cdot(\vec{r}_1-\vec{r}_2)} U(\vec{r}_1 - \vec{r}_2) \frac{1}{V} e^{i\vec{k}'\cdot(\vec{r}_1-\vec{r}_2)} d\vec{r}_1 d\vec{r}_2 = \frac{1}{N\Omega} \int e^{-i(\vec{k}-\vec{k}')\cdot\vec{r}} U(\vec{r}) d\vec{r} \tag{12}$$

where $\vec{r} = \vec{r}_1 - \vec{r}_2$ is the relative coordinate and $V = N\Omega$ is the total volume.

By substituting the expansion of the wavefunction into the Schrödinger equation and projecting onto the plane-wave basis, we obtain an integral equation for the coefficients $g(\vec{k})$:

$$(2E_k - E)g(\vec{k}) + \sum_{\vec{k}'} U_{\vec{k}\vec{k}'} g(\vec{k}') = 0 \tag{13}$$

restricted to the range $E_F < E_k, E_k' < E_F + \hbar\omega_D$. To solve Eq. 13, we consider the **approximation where the scattering amplitudes are constant within the interaction shell**, $U_{\vec{k}\vec{k}'} = -U_0/N$, with $U_0 > 0$. The equation then simplifies to:

$$(2E_k - E)g(\vec{k}) - \frac{U_0}{N} \sum_{\vec{k}'} g(\vec{k}') = 0 \tag{14}$$

Analysis of Eq. 14 reveals that for triplet states, where $g(\vec{k})$ is an odd function, the summation $\sum g(\vec{k}')$ vanishes identically, leaving the triplet energies unaffected by U_0. Conversely, for singlet states (s-wave pairing), the equation admits a bound-state solution with an energy E lower than $2E_F$. This result is significant as it demonstrates the instability of the conventional Fermi sea. While this specific model favors s-wave pairing, more complex interactions in high-temperature superconductors may favor d-wave symmetry, though we restrict our present focus to the former.

To determine the binding energy $\Delta_b = 2E_F - E_{\text{pair}}$ of the Cooper pair in the bound singlet state ($E_{\text{pair}} < 2E_F$), we set $E = E_{\text{pair}}$ in the simplified Schrödinger equation. By dividing the expression by $(2E_k - E_{\text{pair}})$—a quantity that remains positive within the defined energy shell—and summing over all allowed wavevectors $\vec{k}$, we obtain the integral compatibility condition:

$$1 = \frac{U_0}{N} \sum_{\vec{k}} \frac{1}{2E_k - E_{\text{pair}}} \tag{15}$$

where the summation is restricted to the range $E_F < E_k < E_F + \hbar\omega_D$. This discrete sum can be transformed into an energy integral as follows:

$$1 = \frac{U_0}{N} \int_{E_F}^{E_F+\hbar\omega_D} \frac{D_0(E)}{2E - E_{\text{pair}}} dE \tag{16}$$

In this expression, $D_0(E)$ represents the electronic density-of-states for a single spin direction. Given that the energy shell $\hbar\omega_D$ is much smaller than E_F in most metals, we can neglect the energy dependence of $D_0(E)$ and treat it as a constant evaluated at the Fermi level. We define the density-of-states per unit cell for one spin direction as:

$$n_0 = n_0(E_F) := \frac{D_0(E_F)}{N} \tag{17}$$

Under this approximation, the integral in Eq. 16 becomes:

$$1 = U_0 n_0 \int_{E_F}^{E_F+\hbar\omega_D} \frac{1}{2E - E_{\text{pair}}} dE = \frac{1}{2} U_0 n_0 \ln\left(\frac{2E_F + 2\hbar\omega_D - E_{\text{pair}}}{2E_F - E_{\text{pair}}}\right) \tag{18}$$

Solving for the logarithmic argument yields:

$$\frac{2E_F - E_{\text{pair}}}{2E_F + 2\hbar\omega_D - E_{\text{pair}}} = e^{-\frac{2}{U_0 n_0}} \Leftrightarrow \Delta_b = \frac{\hbar\omega_D e^{-\frac{1}{U_0 n_0}}}{\sinh\left(\frac{1}{U_0 n_0}\right)} \tag{19}$$

In the weak-coupling limit ($U_0 n_0 \ll 1$), where $\Delta_b \ll \hbar\omega_D$, the binding energy simplifies to:

$$\Delta_b = 2\hbar\omega_D e^{-\frac{2}{U_0 n_0}} \tag{20}$$

The result in Eq. 20 highlights several key physical insights. First, the free-electron gas is inherently unstable ($\Delta_b > 0$) regardless of how small U_0 might be, leading to the formation of singlet s-wave Cooper pairs. Second, the binding energy is a non-analytic function of U_0 at the origin, meaning it cannot be expanded in a Taylor series. This explains why conventional perturbation theory fails to describe the superconducting transition.

Furthermore, **this model suggests that superconductivity is more prevalent in materials with high U_0 values, which corresponds to strong electron-phonon coupling and, consequently, high resistivity in the normal state**. This explains why "poor" conductors like lead are superconductors at relatively high temperatures, while "excellent" conductors such as copper or gold do not exhibit superconductivity under normal conditions. Finally, the singlet pairing mechanism ($\vec{k}\uparrow, -\vec{k}\downarrow$) is highly sensitive to magnetic impurities, which act to break the pairs and drastically depress the critical temperature.

Single Cooper pair with the second quantization formalism.

Before concluding this section, it is instructive to describe the single Cooper pair using the second quantization formalism. This treatment of a "two-electron" pair within the many-body apparatus serves as a useful algebraic exercise before considering the full many-body theory of superconductivity, allowing for a better visualization of the physical facts beyond the formalism.

We note that the spatial wavefunction, using the symmetry property $g(\vec{k}) = g(-\vec{k})$, can be recast in the form:

$$\psi(\vec{r}_1\sigma_1, \vec{r}_2\sigma_2) = \sum_{\vec{k}} g(\vec{k}) \frac{1}{\sqrt{2}} \frac{1}{V} \left[e^{i\vec{k}\cdot(\vec{r}_1-\vec{r}_2)} \alpha(1)\beta(2) - e^{-i\vec{k}\cdot(\vec{r}_1-\vec{r}_2)} \beta(1)\alpha(2) \right] \tag{21}$$

which is equivalent to the second quantization expression:

$$\psi(\vec{r}_1\sigma_1, \vec{r}_2\sigma_2) = \sum_{\vec{k}} g(\vec{k}) \mathcal{A}\{W_{\vec{k}\uparrow} W_{-\vec{k}\downarrow}\} = \sum_{\vec{k}} g(\vec{k}) c^\dagger_{\vec{k}\uparrow} c^\dagger_{-\vec{k}\downarrow} |\text{Vac}\rangle \tag{22}$$

where $\mathcal{A}$ denotes the determinantal state, $|\text{Vac}\rangle$ is the vacuum state, and $c^\dagger_{\vec{k}\uparrow}$ creates an electron with momentum $\hbar\vec{k}$ and spin-up.

The normal ground state $|\Psi_N\rangle$ of a free-electron metal is represented as a Slater determinant formed by spin-orbitals with wavevectors up to k_F. In the second quantization formalism, we write:

$$|\Psi_N\rangle = \prod_{k<k_F} c^\dagger_{\vec{k}\uparrow} c^\dagger_{\vec{k}\downarrow} |\text{Vac}\rangle \quad \text{or equivalently} \quad |\Psi_N\rangle = \prod_{k<k_F} c^\dagger_{\vec{k}\uparrow} c^\dagger_{-\vec{k}\downarrow} |\text{Vac}\rangle \tag{23}$$

Choosing to create electrons in pairs $(\vec{k}\uparrow, -\vec{k}\downarrow)$ facilitates the forthcoming elaborations. For a single Cooper pair added to the system with the restriction $g(\vec{k}) = 0$ for $E_k < E_F$, the wavefunction is:

$$|\Psi_{\text{Cooper pair}}\rangle = \sum_{\vec{k}} g(\vec{k}) c^\dagger_{\vec{k}\uparrow} c^\dagger_{-\vec{k}\downarrow} |\Psi_N\rangle \tag{24}$$

This indicates that the two additional electrons have total spin and total momentum equal to zero. The Hamiltonian for this system of interacting fermions in second quantized form is:

$$H = \sum_{\vec{k}} E_k \left(c^\dagger_{\vec{k}\uparrow} c_{\vec{k}\uparrow} + c^\dagger_{-\vec{k}\downarrow} c_{-\vec{k}\downarrow} \right) + \sum_{\vec{k},\vec{k}'} U_{\vec{k}\vec{k}'} c^\dagger_{\vec{k}\uparrow} c^\dagger_{-\vec{k}\downarrow} c_{-\vec{k}'\downarrow} c_{\vec{k}'\uparrow} \tag{25}$$

The first term represents the kinetic energy of independent fermions, while the second accounts for the pairing interaction. The creation and annihilation operators satisfy the standard anticommutation rules:

$$\{c_{\vec{k}\sigma}, c_{\vec{k}'\sigma'}\} = \{c^{\dagger}_{\vec{k}\sigma}, c^{\dagger}_{\vec{k}'\sigma'}\} = 0; \quad \{c_{\vec{k}\sigma}, c^{\dagger}_{\vec{k}'\sigma'}\} = \delta_{\vec{k}\vec{k}'}\delta_{\sigma\sigma'} \tag{26}$$

From these equations, one can re-derive the results for the isolated Cooper pair, specifically the compatibility equations. The next theoretical step involves allowing the simultaneous pairing of all electrons in the Fermi sea.

3. Ground-state at zero temperature in the BCS theory.

3.1 Variational determination of the ground-state wavefunction.

The Cooper model (1956) provides an illuminating point of departure for the theory of superconductivity, leading to the microscopic breakthrough by Bardeen, Cooper, and Schrieffer (BCS) in 1957. While the Cooper treatment demonstrates that the normal electron gas is unstable under a net attractive interaction for even a single pair, **a realistic theory must describe a cooperative condensation minimizing the total energy at** $T = 0$. This is achieved via the BCS variational ground-state wavefunction:

$$|\Psi_S\rangle = \prod_{\vec{k}} \left(u_{\vec{k}} + v_{\vec{k}} c^{\dagger}_{\vec{k}\uparrow} c^{\dagger}_{-\vec{k}\downarrow} \right) |\text{Vac}\rangle \tag{27}$$

where $|\text{Vac}\rangle$ is the vacuum, while $u_{\vec{k}}$ and $v_{\vec{k}}$ are real, even functions of $\vec{k}$. Under the normalization constraint $u^2_{\vec{k}} + v^2_{\vec{k}} = 1$, $v_{\vec{k}}$ and $u_{\vec{k}}$ represent probability amplitudes for the pair $(\vec{k}\uparrow, -\vec{k}\downarrow)$ being occupied or unoccupied. The normal metal state is a specific case of this wavefunction defined by:

$$\begin{cases} u_{\vec{k}} = 0, v_{\vec{k}} = 1 & \text{for } k < k_F \\ u_{\vec{k}} = 1, v_{\vec{k}} = 0 & \text{for } k > k_F \end{cases} \tag{28}$$

Since the total number of electrons is not fixed in $|\Psi_S\rangle$, we minimize $\langle\Psi_S|H|\Psi_S\rangle$ under the constraint $\langle\Psi_S|N_{\text{op}}|\Psi_S\rangle = N$, where the particle number operator is:

$$N_{\text{op}} = \sum_{\vec{k}} \left(c^{\dagger}_{\vec{k}\uparrow} c_{\vec{k}\uparrow} + c^{\dagger}_{-\vec{k}\downarrow} c_{-\vec{k}\downarrow} \right) \tag{29}$$

Applying a Lagrange multiplier μ (the Fermi energy), we minimize $W_S = \langle\Psi_S|H_{BCS}|\Psi_S\rangle$ using the BCS Hamiltonian:

$$H_{BCS} = \sum_{\vec{k}} \xi_{\vec{k}} \left(c^{\dagger}_{\vec{k}\uparrow} c_{\vec{k}\uparrow} + c^{\dagger}_{-\vec{k}\downarrow} c_{-\vec{k}\downarrow} \right) + \sum_{\vec{k}\vec{k}'} V_{\vec{k}\vec{k}'} c^{\dagger}_{\vec{k}\uparrow} c^{\dagger}_{-\vec{k}\downarrow} c_{-\vec{k}'\downarrow} c_{\vec{k}'\uparrow} \tag{30}$$

where $\xi_{\vec{k}} = \frac{\hbar^2 k^2}{2m} - \mu$. Evaluation of the expectation value yields the following matrix elements:

$$\langle 0|(u_{\vec{k}} + v_{\vec{k}} c_{-\vec{k}\downarrow} c_{\vec{k}\uparrow})(u_{\vec{k}} + v_{\vec{k}} c^\dagger_{\vec{k}\uparrow} c^\dagger_{-\vec{k}\downarrow})|0\rangle = u^2_{\vec{k}} + v^2_{\vec{k}} \tag{31}$$

$$\langle \Psi_S|\Psi_S\rangle = \prod_{\vec{k}} (u^2_{\vec{k}} + v^2_{\vec{k}}) = 1 \quad \forall \vec{k} \tag{32}$$

$$\langle \Psi_S|c^\dagger_{\vec{k}\uparrow} c_{\vec{k}\uparrow}|\Psi_S\rangle = \langle 0|(u_{\vec{k}} + v_{\vec{k}} c_{-\vec{k}\downarrow} c_{\vec{k}\uparrow}) c^\dagger_{\vec{k}\uparrow} c_{\vec{k}\uparrow} (u_{\vec{k}} + v_{\vec{k}} c^\dagger_{\vec{k}\uparrow} c^\dagger_{-\vec{k}\downarrow})|0\rangle = v^2_{\vec{k}} \tag{33}$$

$$\langle \Psi_S|c^\dagger_{\vec{k}\uparrow} c^\dagger_{-\vec{k}\downarrow}|\Psi_S\rangle = \langle \Psi_S|c_{-\vec{k}\downarrow} c_{\vec{k}\uparrow}|\Psi_S\rangle = u_{\vec{k}} v_{\vec{k}} \tag{34}$$

By applying these matrix elements, the expectation value for the superconductor ground-state energy is expressed as:

$$W_S = \langle \Psi_S|H_{BCS}|\Psi_S\rangle = 2\sum_{\vec{k}} \xi_{\vec{k}} v^2_{\vec{k}} + \sum_{\vec{k}\vec{k}'} V_{\vec{k}\vec{k}'} u_{\vec{k}} v_{\vec{k}} u_{\vec{k}'} v_{\vec{k}'} \tag{35}$$

The task reduces to a standard algebraic minimization of W_S with respect to the variational parameters $u_{\vec{k}}$ and $v_{\vec{k}}$ in reciprocal space. To incorporate the normalization constraint $u^2_{\vec{k}} + v^2_{\vec{k}} = 1$ naturally, we introduce a polar representation:

$$\begin{cases} u_{\vec{k}} = \cos\theta_{\vec{k}} \\ v_{\vec{k}} = \sin\theta_{\vec{k}} \end{cases} \Rightarrow \sin 2\theta_{\vec{k}} = 2u_{\vec{k}} v_{\vec{k}}; \quad \cos 2\theta_{\vec{k}} = u^2_{\vec{k}} - v^2_{\vec{k}} \tag{36}$$

Substituting these into the energy expression yields:

$$W_S = 2\sum_{\vec{k}} \xi_{\vec{k}} \sin^2\theta_{\vec{k}} + \frac{1}{4}\sum_{\vec{k}\vec{k}'} V_{\vec{k}\vec{k}'} \sin 2\theta_{\vec{k}} \sin 2\theta_{\vec{k}'} \tag{37}$$

Setting the partial derivative with respect to $\theta_{\vec{k}}$ to zero for the minimization condition gives:

$$\begin{aligned} \frac{\partial W_S}{\partial \theta_{\vec{k}}} = 0 \Leftrightarrow 2\xi_{\vec{k}} \sin 2\theta_{\vec{k}} + \left(\sum_{\vec{k}'} V_{\vec{k}\vec{k}'} \sin 2\theta_{\vec{k}'}\right) \cos 2\theta_{\vec{k}} = 0 \Leftrightarrow \\ \Leftrightarrow 2\xi_{\vec{k}} u_{\vec{k}} v_{\vec{k}} + \left(\sum_{\vec{k}'} V_{\vec{k}\vec{k}'} u_{\vec{k}'} v_{\vec{k}'}\right)(u^2_{\vec{k}} - v^2_{\vec{k}}) = 0 \quad \forall \vec{k} \end{aligned} \tag{38}$$

To solve this self-consistent set of equations, we define the energy gap parameter $\Delta_{\vec{k}}$ as:

$$\Delta_{\vec{k}} = -\sum_{\vec{k}'} V_{\vec{k}\vec{k}'} u_{\vec{k}'} v_{\vec{k}'} \tag{39}$$

The variational condition is then simplified to:

$$2\xi_{\vec{k}} u_{\vec{k}} v_{\vec{k}} - \Delta_{\vec{k}}(u_{\vec{k}}^2 - v_{\vec{k}}^2) = 0 \tag{40}$$

Solving this alongside the normalization condition $u_{\vec{k}}^2 + v_{\vec{k}}^2 = 1$ leads to the following expressions for the coherence factors:

$$u_{\vec{k}}^2 = \frac{1}{2}\left(1 + \frac{\xi_{\vec{k}}}{\sqrt{\xi_{\vec{k}}^2 + \Delta_{\vec{k}}^2}}\right); \quad v_{\vec{k}}^2 = \frac{1}{2}\left(1 - \frac{\xi_{\vec{k}}}{\sqrt{\xi_{\vec{k}}^2 + \Delta_{\vec{k}}^2}}\right) \tag{41}$$

The signs are chosen to ensure that as the interaction $V_{\vec{k}\vec{k}'}$ (and thus $\Delta_{\vec{k}}$) vanishes, we recover the distribution of the normal ground state. Finally, substituting $u_{\vec{k}} v_{\vec{k}}$ back into the definition of the gap results in the BCS self-consistent gap equation:

$$\Delta_{\vec{k}} = -\frac{1}{2}\sum_{\vec{k}'} V_{\vec{k}\vec{k}'} \frac{\Delta_{\vec{k}'}}{\sqrt{\xi_{\vec{k}'}^2 + \Delta_{\vec{k}'}^2}} \quad \forall \vec{k} \tag{42}$$

A trivial solution to Eq. 42 is given by $\Delta_{\vec{k}} = 0 \quad \forall \vec{k}$, which corresponds to the normal metal state where electrons occupy the Fermi sphere up to k_F. Generally, the integral equation 42 presents significant analytical difficulties. For simplicity, and following the methodology established for a single s-wave Cooper pair, the matrix elements $V_{\vec{k}\vec{k}'}$ are treated as constant within a specific energy shell surrounding the Fermi level. In this average potential approximation, we define:

$$V_{\vec{k}\vec{k}'} = \begin{cases} -\frac{U_0}{N} & \text{if } |\xi_{\vec{k}}|, |\xi_{\vec{k}'}| < \hbar\omega_D \quad (U_0 > 0) \\ 0 & \text{otherwise} \end{cases} \tag{43}$$

where U_0 is a positive constant independent of the number of unit cells N. From this approximation and Eq. 39, it follows that:

$$\Delta_{\vec{k}} = \begin{cases} \Delta_0 & \text{if } |\xi_{\vec{k}}| < \hbar\omega_D \\ 0 & \text{otherwise} \end{cases} \tag{44}$$

Consequently, the integral equation 42 simplifies to:

$$1 = \frac{U_0}{2N}\sum_{\vec{k}} \frac{1}{\sqrt{\xi_{\vec{k}}^2 + \Delta_0^2}} \tag{45}$$

under the constraint $-\hbar\omega_D < \xi_{\vec{k}} < \hbar\omega_D$. As is standard, the summation over $\vec{k}$ is converted into an integral. By assuming the electron density-of-states is constant in the narrow energy shell around the Fermi level, we obtain:

$$1 = \frac{1}{2} U_0 n_0 \int_{-\hbar\omega_D}^{\hbar\omega_D} \frac{d\xi}{\sqrt{\xi^2 + \Delta_0^2}} \tag{46}$$

where n_0 represents the density-of-states for one spin direction per unit cell at the Fermi energy. The calculation proceeds using the indefinite integral:

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \sinh^{-1} \frac{x}{|a|} \Rightarrow 1 = U_0 n_0 \sinh^{-1} \frac{\hbar\omega_D}{\Delta_0} \tag{47}$$

which yields:

$$\Delta_0 = \frac{\hbar\omega_D}{\sinh(1/U_0 n_0)} \tag{48}$$

In the weak coupling limit, where $U_0 n_0 \ll 1$, the expression results in:

$$\Delta_0 = 2\hbar\omega_D e^{-\frac{1}{U_0 n_0}} \tag{49}$$

For typical conventional superconductors, $E_F \sim 1$ eV and $U_0 \sim 0.1 - 0.5$ eV, leading to a dimensionless coupling parameter $U_0 n_0 \sim \frac{U_0}{E_F} \sim 0.1 - 0.5$. Thus, $\Delta_0 \sim 1$ meV is generally a small fraction of $\hbar\omega_D$. While this BCS framework assumes weak coupling and s-wave symmetry, **it has been generalized for strong-coupling scenarios by McMillan and Eliashberg**. Furthermore, materials like magnesium diboride (MgB_2) exhibit two electronic bands and two gaps. High-T_c materials, including cuprates and iron pnictides, present a major challenge to the original formulation as the pairing mechanisms are more complex, potentially involving electronic correlations and unconventional non-zero angular momentum states.

3.2 Ground-state energy and isotope effect.

The condensation energy of a superconductor is defined as the energy difference between the superconducting ground-state energy W_S and the normal state energy W_N:

$$W_S - W_N = 2\sum_{\vec{k}} \xi_{\vec{k}} v_{\vec{k}}^2 + \sum_{\vec{k}\vec{k}'} V_{\vec{k}\vec{k}'} u_{\vec{k}} v_{\vec{k}} u_{\vec{k}'} v_{\vec{k}'} - 2\sum_{k<k_F} \xi_{\vec{k}} \tag{50}$$

By substituting the gap parameter definition from Eq. 39 and the expressions for the coherence factors from Eq. 41, the energy difference is rewritten as:

$$\begin{aligned} W_S - W_N &= \sum_{\vec{k}} (2\xi_{\vec{k}} v_{\vec{k}}^2 - \Delta_{\vec{k}} u_{\vec{k}} v_{\vec{k}}) - \sum_{k<k_F} 2\xi_{\vec{k}} = \\ &= \sum_{\vec{k}} \left(\xi_{\vec{k}} - \frac{\xi_{\vec{k}}^2}{\sqrt{\xi_{\vec{k}}^2 + \Delta_{\vec{k}}^2}} - \frac{1}{2} \frac{\Delta_{\vec{k}}^2}{\sqrt{\xi_{\vec{k}}^2 + \Delta_{\vec{k}}^2}} \right) - \sum_{k<k_F} 2\xi_{\vec{k}} \end{aligned} \tag{51}$$

Under the average potential approximation, where $\Delta_{\vec{k}} = \Delta_0$ for $|\xi_{\vec{k}}| < \hbar\omega_D$, we convert the sum into an integral. **Assuming a constant density-of-states** $D_0(E_F)$ **at the Fermi level**, the expression becomes:

$$W_S - W_N = D_0(E_F) \int_{-\hbar\omega_D}^{\hbar\omega_D} \left(\xi - \frac{2\xi^2 + \Delta_0^2}{2\sqrt{\xi^2 + \Delta_0^2}} \right) d\xi - D_0(E_F) \int_{-\hbar\omega_D}^{0} 2\xi d\xi \tag{52}$$

Evaluating this using the indefinite integral $\int \frac{2x^2+a^2}{\sqrt{x^2+a^2}} dx = x\sqrt{x^2+a^2}$, we obtain:

$$W_S - W_N = D_0(E_F) \left(-\hbar\omega_D \sqrt{\hbar^2\omega_D^2 + \Delta_0^2} + \hbar^2\omega_D^2 \right) \tag{53}$$

In the weak binding limit where $\Delta_0 \ll \hbar\omega_D$, a Taylor expansion of the square root yields the final expression for the condensation energy:

$$W_S - W_N = -\frac{1}{2} D_0(E_F) \Delta_0^2 \tag{54}$$

This result indicates that the energy reduction arises from electrons within the energy shell Δ_0 around the Fermi level, which lower their energy through the pairing mechanism (see Fig. 6).

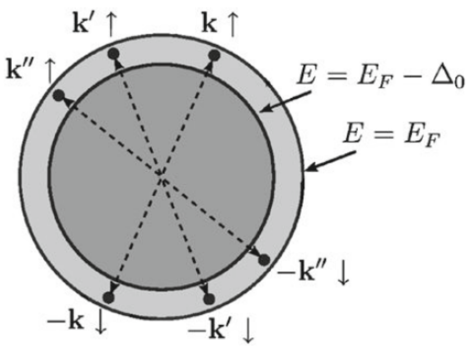


Figure 6: Schematic representation of the origin of the condensation energy in the BCS ground state. Electron pairs in the blurred region of k-space, within an energy shell Δ_0 around the Fermi level, contribute to the condensation energy. Typically, $\Delta_0 \sim 1$ meV while $E_F \sim 1$ eV.

The thermodynamic critical magnetic field $H_c(0)$ at absolute zero is fundamentally linked to the condensation energy per unit volume through the following relation:

$$\frac{1}{V}(W_N - W_S) = \frac{H_c^2(0)}{8\pi} = \frac{1}{2} D_0(E_F) \Delta_0^2 \frac{1}{V} \Rightarrow H_c^2(0) = \frac{4\pi n_0 \Delta_0^2}{\Omega} \tag{55}$$

In this expression, n_0 denotes the density-of-states at the Fermi level for a single spin direction per unit cell as defined in 17, while Ω represents the volume of the unit cell.

To approximate the magnitude of $H_c(0)$, we consider the relationship:

$$\mu_B^2 H_c^2(0) = 4\pi \frac{\Delta_0^2}{E_F} \frac{\mu_B^2}{(10a_B)^3} \tag{56}$$

By adopting typical values such as $\mu_B \simeq 5.788 \cdot 10^{-6}$ meV/gauss, $\Delta_0 \sim 1$ meV, $E_F \sim 10^3$ meV, and the ratio $\frac{\mu_B^2}{a_B^3} \simeq 0.363$ meV, the estimated critical field $H_c(0)$ scales to several hundred or a few thousand gauss.

Furthermore, the formulation for the critical field **inherently accounts for the isotope effect, which historically provided the primary evidence for phonon-mediated electron-electron coupling**. For metals composed of different isotopes with mass M, the substitution primarily affects the Debye frequency $\hbar\omega_D$, while electronic parameters remain largely invariant. Given that $\hbar\omega_D \propto 1/\sqrt{M}$ in the Debye model, and applying the relations for Δ_0 and $H_c(0)$, we arrive at the following scaling law:

$$H_c(0) \cdot M^\alpha = \text{const} \tag{57}$$

Within the weak coupling limit, the theory predicts $\alpha = 0.5$. While this value is accurately observed in non-transition metals like mercury, the experimental reality is more diverse. Metals such as Mo, Re, and Os exhibit $\alpha < 0.5$, whereas in Ru and Zr, the isotope effect is virtually absent. Such deviations, particularly prominent in high-T_c superconductors, emphasize that the current model relies on simplifying assumptions that must be tailored to the specific mediation mechanisms of real materials.

3.3 Momentum distribution and coherence length.

The probability of occupying a state with momentum $\vec{k}$ and spin σ in the superconducting ground state is determined by the expectation value of the single-particle number operator:

$$\langle \Psi_S | c^\dagger_{\vec{k}\sigma} c_{\vec{k}\sigma} | \Psi_S \rangle = v_{\vec{k}}^2 = \frac{1}{2}\left(1 - \frac{\xi_{\vec{k}}}{\sqrt{\xi_{\vec{k}}^2 + \Delta_{\vec{k}}^2}}\right) \tag{58}$$

In contrast, for the normal electron gas, this occupancy probability follows the standard step-function distribution at $T = 0$:

$$\langle \Psi_N | c^\dagger_{\vec{k}\sigma} c_{\vec{k}\sigma} | \Psi_N \rangle = \begin{cases} 1 & \text{for } k < k_F \\ 0 & \text{for } k > k_F \end{cases} \tag{59}$$

The transition from the sharp Fermi surface to the smooth distribution in the superconducting state is characterized by the coherence factors. While $v_{\vec{k}}^2$ represents the occupation probability, the product $u_{\vec{k}} v_{\vec{k}}$ vanishes in the normal state but becomes significant near the Fermi level in the superconductor, peaking where $\xi_{\vec{k}} = 0$. These behaviors are schematically compared in Fig. 7.

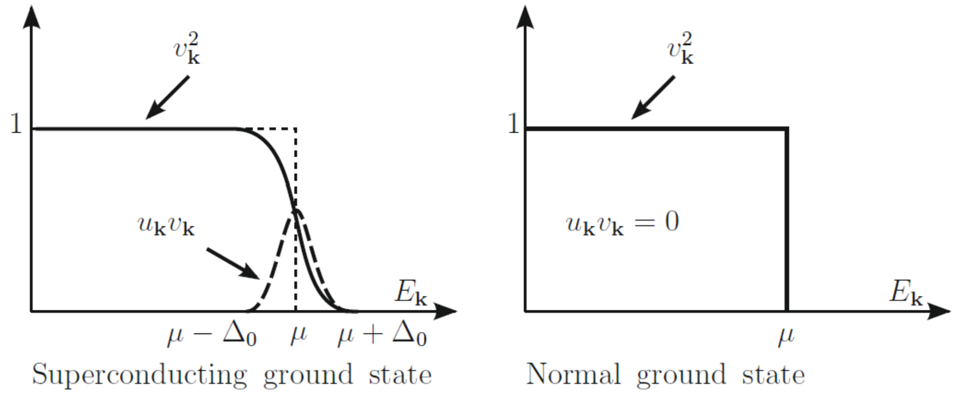


Figure 7: Schematic behavior of the quantities $v_{\vec{k}}^2$ and $u_{\vec{k}}v_{\vec{k}}$ for both the superconducting and normal ground states. In the superconductor, the distribution $v_{\vec{k}}^2$ is smeared over an energy range determined by the gap Δ_0 around the chemical potential μ.

The pair operator $c_{\vec{k}\uparrow}^\dagger c_{-\vec{k}\downarrow}^\dagger$ (or its adjoint $c_{-\vec{k}\downarrow}c_{\vec{k}\uparrow}$) characterizes the superconducting order. In the superconducting ground state, the expectation value of this operator is given by:

$$\langle \Psi_S | c_{\vec{k}\uparrow}^\dagger c_{-\vec{k}\downarrow}^\dagger | \Psi_S \rangle = u_{\vec{k}}v_{\vec{k}} = \frac{1}{2}\frac{\Delta_{\vec{k}}}{\sqrt{\xi_{\vec{k}}^2 + \Delta_{\vec{k}}^2}} \tag{60}$$

While this quantity is identically zero in the normal state, it remains finite within a narrow shell around k_F in the superconducting state. **This non-zero coherence is the microscopic origin of fundamental phenomena such as the Meissner effect and Josephson tunneling**.

The region in reciprocal space where $u_{\vec{k}}v_{\vec{k}}$ is significant spans an energy width Δ_0 around the Fermi level. The corresponding momentum width δk around the Fermi wavevector is estimated as:

$$\frac{\hbar^2}{2m}\left((k_F \pm \delta k)^2 - k_F^2\right) \simeq \pm\Delta_0 \Rightarrow \frac{\hbar^2 k_F}{m}\delta k \simeq \Delta_0 \Rightarrow \delta k \simeq \frac{\Delta_0}{\hbar v_F} \tag{61}$$

where v_F denotes the Fermi velocity. According to the uncertainty principle, **the spatial extent of the Cooper pair, known as the BCS coherence length** ξ_0, is estimated by $\xi_0 \simeq \frac{1}{\delta k}$:

$$\xi_0 = \frac{1}{\pi}\frac{\hbar v_F}{\Delta_0} = \frac{1}{\pi}\frac{\hbar^2 k_F}{m\Delta_0} \tag{62}$$

The numerical factor π arises from a more rigorous analysis. The length ξ_0 represents the average spatial separation between the two electrons constituting a pair. Typical values for ξ_0 range from hundreds to thousands of Å in conventional superconductors, decreasing to tens of Å in high-T_c materials where the BCS framework requires modification.

In conventional superconductors, a vast number of Cooper pairs occupy the volume defined by the coherence length ξ_0. The density of Cooper pairs is $\sim n\frac{\Delta_0}{E_F}$, where n is the total electron density. The average distance d between the centers of mass of the pairs relates to the inter-electronic distance $r_s a_B$ as $d \simeq r_s a_B \left(\frac{E_F}{\Delta_0}\right)^{\frac{1}{3}}$. For $E_F \sim 1$ eV and $\Delta_0 \sim 1$ meV, $d \sim 10$ Å. Given $\xi_0 \sim 1000$ Å, millions of Cooper pairs overlap within the same spatial region, emphasizing that the binding energy $2\Delta_0$ of any single pair is a collective property dependent on the entire ensemble.

4. Excited states at zero temperature.

4.1 The Bogoliubov canonical transformation.

While our previous analysis focused on the superconducting ground state at $T = 0$, the study of excited states is essential for understanding the thermal and transport properties of the system. The original BCS approach involved applying creation and annihilation operators to the ground-state wavefunction. However, a more computationally efficient framework was provided by the canonical transformations of Bogoliubov and Valatin (1958), which **diagonalize the Hamiltonian to reveal the quasiparticle excitation spectrum**.

We consider the BCS Hamiltonian as defined in 30:

$$H_{BCS} = \sum_{\vec{k}\sigma} \xi_{\vec{k}} c^\dagger_{\vec{k}\sigma} c_{\vec{k}\sigma} + \sum_{\vec{k}\vec{k}'} V_{\vec{k}\vec{k}'} c^\dagger_{\vec{k}\uparrow} c^\dagger_{-\vec{k}\downarrow} c_{-\vec{k}'\downarrow} c_{\vec{k}'\uparrow} \tag{63}$$

The ground-state wavefunction is expressed in the variational form:

$$|\Psi_S\rangle = \prod_{\vec{k}} \left(u_{\vec{k}} + v_{\vec{k}} c^\dagger_{\vec{k}\uparrow} c^\dagger_{-\vec{k}\downarrow} \right) |\mathrm{Vac}\rangle \tag{64}$$

In the Bogoliubov framework, the real coefficients $u_{\vec{k}}$ and $v_{\vec{k}}$ (satisfying $u^2_{\vec{k}} + v^2_{\vec{k}} = 1$) are determined such that the Hamiltonian is diagonalized. This is achieved by defining four new fermion operators $\gamma_{\vec{k}\sigma}$ and $\gamma^\dagger_{\vec{k}\sigma}$ through a linear canonical transformation that mixes creation and annihilation operators of paired spin-orbitals:

$$\begin{pmatrix} \gamma_{\vec{k}\uparrow} \\ \gamma^\dagger_{-\vec{k}\downarrow} \end{pmatrix} = \begin{pmatrix} u_{\vec{k}} & -v_{\vec{k}} \\ v_{\vec{k}} & u_{\vec{k}} \end{pmatrix} \begin{pmatrix} c_{\vec{k}\uparrow} \\ c^\dagger_{-\vec{k}\downarrow} \end{pmatrix} ; \quad \begin{pmatrix} \gamma^\dagger_{\vec{k}\uparrow} \\ \gamma_{-\vec{k}\downarrow} \end{pmatrix} = \begin{pmatrix} u_{\vec{k}} & -v_{\vec{k}} \\ v_{\vec{k}} & u_{\vec{k}} \end{pmatrix} \begin{pmatrix} c^\dagger_{\vec{k}\uparrow} \\ c_{-\vec{k}\downarrow} \end{pmatrix} \tag{65}$$

The explicit form of these Bogoliubov transformations is:

$$\begin{cases} \gamma_{\vec{k}\uparrow} = u_{\vec{k}} c_{\vec{k}\uparrow} - v_{\vec{k}} c^\dagger_{-\vec{k}\downarrow} \\ \gamma^\dagger_{-\vec{k}\downarrow} = v_{\vec{k}} c_{\vec{k}\uparrow} + u_{\vec{k}} c^\dagger_{-\vec{k}\downarrow} \end{cases} ; \quad \begin{cases} \gamma^\dagger_{\vec{k}\uparrow} = u_{\vec{k}} c^\dagger_{\vec{k}\uparrow} - v_{\vec{k}} c_{-\vec{k}\downarrow} \\ \gamma_{-\vec{k}\downarrow} = v_{\vec{k}} c^\dagger_{\vec{k}\uparrow} + u_{\vec{k}} c_{-\vec{k}\downarrow} \end{cases} \tag{66}$$

The corresponding inverse transformations are given by:

$$\begin{cases} c_{\vec{k}\uparrow} = u_{\vec{k}}\gamma_{\vec{k}\uparrow} + v_{\vec{k}}\gamma^\dagger_{-\vec{k}\downarrow} \\ c^\dagger_{-\vec{k}\downarrow} = u_{\vec{k}}\gamma^\dagger_{-\vec{k}\downarrow} - v_{\vec{k}}\gamma_{\vec{k}\uparrow} \end{cases} ; \quad \begin{cases} c^\dagger_{\vec{k}\uparrow} = u_{\vec{k}}\gamma^\dagger_{\vec{k}\uparrow} + v_{\vec{k}}\gamma_{-\vec{k}\downarrow} \\ c_{-\vec{k}\downarrow} = u_{\vec{k}}\gamma_{-\vec{k}\downarrow} - v_{\vec{k}}\gamma^\dagger_{\vec{k}\uparrow} \end{cases} \tag{67}$$

These transformations are canonical because the γ operators preserve the standard fermion anticommutation relations. In the limit of a normal metal, these operators simplify to represent free electron and hole excitations. However, in the superconducting state, **they describe the creation and annihilation of quasiparticles within the correlated electron system**, where the attractive pairing interaction is active.

The utility of the Bogoliubov transformations becomes evident when considering the effect of the annihilation operators on the BCS ground state. By applying $\gamma_{\vec{k}\uparrow}$ to the variational form, we observe:

$$\begin{aligned} \gamma_{\vec{k}\uparrow}(u_{\vec{k}} + v_{\vec{k}}c^\dagger_{\vec{k}\uparrow}c^\dagger_{-\vec{k}\downarrow}) &= (u_{\vec{k}}c_{\vec{k}\uparrow} - v_{\vec{k}}c^\dagger_{-\vec{k}\downarrow})(u_{\vec{k}} + v_{\vec{k}}c^\dagger_{\vec{k}\uparrow}c^\dagger_{-\vec{k}\downarrow}) = \\ &= (u^2_{\vec{k}} + u_{\vec{k}}v_{\vec{k}}c^\dagger_{\vec{k}\uparrow}c^\dagger_{-\vec{k}\downarrow})c_{\vec{k}\uparrow} \end{aligned} \tag{68}$$

This identity implies that:

$$\gamma_{\vec{k}\uparrow}|\Psi_S\rangle = \gamma_{\vec{k}\downarrow}|\Psi_S\rangle = 0 \tag{69}$$

Thus, the BCS wavefunction acts as the vacuum state for the Bogoliubov quasiparticles, regardless of the specific values of $u_{\vec{k}}$ and $v_{\vec{k}}$.

To determine the quasiparticle energies, we address the quartic terms in the BCS Hamiltonian. Following mean-field theory, we decompose the product of creation operators into its expectation value $a_{\vec{k}}$ and a fluctuation term:

$$c^\dagger_{\vec{k}\uparrow}c^\dagger_{-\vec{k}\downarrow} = a_{\vec{k}} + \left(c^\dagger_{\vec{k}\uparrow}c^\dagger_{-\vec{k}\downarrow} - a_{\vec{k}}\right) \tag{70}$$

where $a_{\vec{k}}$ is defined as the ground-state expectation value:

$$a_{\vec{k}} = \langle c^\dagger_{\vec{k}\uparrow}c^\dagger_{-\vec{k}\downarrow}\rangle \tag{71}$$

Assuming fluctuations are small, we linearize the Hamiltonian by neglecting second-order terms in the fluctuation operators. Substituting these expressions into H_{BCS} results in the simplified Bogoliubov Hamiltonian H_B:

$$H_B = \sum_{\vec{k}\sigma}\xi_{\vec{k}}c^\dagger_{\vec{k}\sigma}c_{\vec{k}\sigma} + \sum_{\vec{k}\vec{k}'}V_{\vec{k}\vec{k}'}\left(a_{\vec{k}}c_{-\vec{k}'\downarrow}c_{\vec{k}'\uparrow} + a_{\vec{k}'}c^\dagger_{\vec{k}\uparrow}c^\dagger_{-\vec{k}\downarrow} - a_{\vec{k}}a_{\vec{k}'}\right) \tag{72}$$

This effective Hamiltonian is now quadratic in the fermion operators, allowing for explicit diagonalization and the determination of the superconductor's excitation spectrum.

By defining the energy gap parameters $\Delta_{\vec{k}}$ in terms of the expectation values $a_{\vec{k}'}$ as:

$$\Delta_{\vec{k}} = -\sum_{\vec{k}'} V_{\vec{k}\vec{k}'} a_{\vec{k}'} \tag{73}$$

the linearized Bogoliubov Hamiltonian H_B is expressed as:

$$H_B = \sum_{\vec{k}\sigma} \xi_{\vec{k}} c^{\dagger}_{\vec{k}\sigma} c_{\vec{k}\sigma} - \sum_{\vec{k}} \Delta_{\vec{k}} \left(c^{\dagger}_{\vec{k}\uparrow} c^{\dagger}_{-\vec{k}\downarrow} + c_{-\vec{k}\downarrow} c_{\vec{k}\uparrow} \right) + \sum_{\vec{k}} \Delta_{\vec{k}} a_{\vec{k}} \tag{74}$$

To diagonalize this operator, we transform the electron operators into the quasiparticle γ basis using the relations in 67. Substituting these into the kinetic and pairing terms, we obtain:

$$\begin{aligned} c^{\dagger}_{\vec{k}\uparrow} c_{\vec{k}\uparrow} + c^{\dagger}_{-\vec{k}\downarrow} c_{-\vec{k}\downarrow} =& (u^2_{\vec{k}} - v^2_{\vec{k}})(\gamma^{\dagger}_{\vec{k}\uparrow}\gamma_{\vec{k}\uparrow} + \gamma^{\dagger}_{-\vec{k}\downarrow}\gamma_{-\vec{k}\downarrow}) + \\ &+ 2u_{\vec{k}} v_{\vec{k}} (\gamma^{\dagger}_{\vec{k}\uparrow}\gamma^{\dagger}_{-\vec{k}\downarrow} + \gamma_{-\vec{k}\downarrow}\gamma_{\vec{k}\uparrow}) + 2v^2_{\vec{k}} \end{aligned} \tag{75}$$

$$\begin{aligned} c^{\dagger}_{\vec{k}\uparrow} c^{\dagger}_{-\vec{k}\downarrow} + c_{-\vec{k}\downarrow} c_{\vec{k}\uparrow} =& -2u_{\vec{k}} v_{\vec{k}} (\gamma^{\dagger}_{\vec{k}\uparrow}\gamma_{\vec{k}\uparrow} + \gamma^{\dagger}_{-\vec{k}\downarrow}\gamma_{-\vec{k}\downarrow}) + \\ &+ (u^2_{\vec{k}} - v^2_{\vec{k}})(\gamma^{\dagger}_{\vec{k}\uparrow}\gamma^{\dagger}_{-\vec{k}\downarrow} + \gamma_{-\vec{k}\downarrow}\gamma_{\vec{k}\uparrow}) + 2u_{\vec{k}} v_{\vec{k}} \end{aligned} \tag{76}$$

Substituting these expressions back into the Hamiltonian yields the consolidated form:

$$\begin{aligned} H_B =& \sum_{\vec{k}} (\xi_{\vec{k}}(u^2_{\vec{k}} - v^2_{\vec{k}}) + 2\Delta_{\vec{k}} u_{\vec{k}} v_{\vec{k}})(\gamma^{\dagger}_{\vec{k}\uparrow}\gamma_{\vec{k}\uparrow} + \gamma^{\dagger}_{-\vec{k}\downarrow}\gamma_{-\vec{k}\downarrow}) + \\ &+ \sum_{\vec{k}} (2\xi_{\vec{k}} u_{\vec{k}} v_{\vec{k}} - \Delta_{\vec{k}}(u^2_{\vec{k}} - v^2_{\vec{k}}))(\gamma^{\dagger}_{\vec{k}\uparrow}\gamma^{\dagger}_{-\vec{k}\downarrow} + \gamma_{-\vec{k}\downarrow}\gamma_{\vec{k}\uparrow}) + W_0 \end{aligned} \tag{77}$$

where $W_0 = \sum_{\vec{k}} (2\xi_{\vec{k}} v^2_{\vec{k}} - 2\Delta_{\vec{k}} u_{\vec{k}} v_{\vec{k}} + \Delta_{\vec{k}} a_{\vec{k}})$. To ensure H_B is diagonal, the coefficients of the non-conserving terms $\gamma^{\dagger}\gamma^{\dagger}$ and $\gamma\gamma$ must vanish:

$$2\xi_{\vec{k}} u_{\vec{k}} v_{\vec{k}} - \Delta_{\vec{k}} \left(u^2_{\vec{k}} - v^2_{\vec{k}} \right) = 0 \quad \forall \vec{k} \tag{78}$$

This requirement for diagonalization is mathematically identical to the energy minimization condition found in the BCS variational method (Eq. 40). Consequently, the solutions for $u_{\vec{k}}$ and $v_{\vec{k}}$ are those previously determined in Eq. 41, subject to the normalization $u^2_{\vec{k}} + v^2_{\vec{k}} = 1$.

A formal comparison between the Bogoliubov canonical transformation and the BCS variational approach reveals their complete physical equivalence. Specifically, the constant term W_0 in the diagonalized Hamiltonian coincides precisely with the superconducting ground-state energy W_S previously derived in Eq. 35. By substituting the coherence factors $u_{\vec{k}}$ and $v_{\vec{k}}$ from Eq. 41 into the diagonal coefficients of Eq. 77, we find:

$$\xi_{\vec{k}} \left(u^2_{\vec{k}} - v^2_{\vec{k}} \right) + 2\Delta_{\vec{k}} u_{\vec{k}} v_{\vec{k}} = \sqrt{\xi^2_{\vec{k}} + \Delta^2_{\vec{k}}} \tag{79}$$

The linearized Bogoliubov Hamiltonian H_B can therefore be recast into its final diagonal form:

$$H_B = W_S + \sum_{\vec{k}} \omega_{\vec{k}} \left(\gamma^\dagger_{\vec{k}\uparrow} \gamma_{\vec{k}\uparrow} + \gamma^\dagger_{-\vec{k}\downarrow} \gamma_{-\vec{k}\downarrow} \right) \tag{80}$$

where the quasiparticle excitation energies $\omega_{\vec{k}}$ are defined as:

$$\omega_{\vec{k}} = \sqrt{\xi^2_{\vec{k}} + \Delta^2_{\vec{k}}} \tag{81}$$

These $\omega_{\vec{k}}$ represent the **energy required to create a quasiparticle above the collective ground state**. Under the average potential approximation, where the gap parameter $\Delta_{\vec{k}}$ is taken as a constant Δ_0 within the relevant energy shell, the spectrum becomes:

$$\omega_{\vec{k}} = \sqrt{\xi^2_{\vec{k}} + \Delta^2_0} \quad \text{(Superconductor)} \tag{82}$$

In the limit of a vanishing electron-electron interaction ($\Delta_0 = 0$), the spectrum reduces to that of a normal metal:

$$\omega_{\vec{k}} = |\xi_{\vec{k}}| \quad \text{(Normal Metal)} \tag{83}$$

As illustrated in Figure 8, the superconducting state is characterized by a significant modification of the excitation spectrum near the Fermi level. Unlike the normal metal, which allows excitations at arbitrarily low energies, the superconductor exhibits a forbidden energy region. Specifically, there are no available states for electron-like excitations in the interval $(E_F, E_F + \Delta_0)$ or for hole-like excitations in the interval $(E_F - \Delta_0, E_F)$. This energy gap Δ_0 is the hallmark of the superconducting phase and underpins its unique thermodynamic properties.

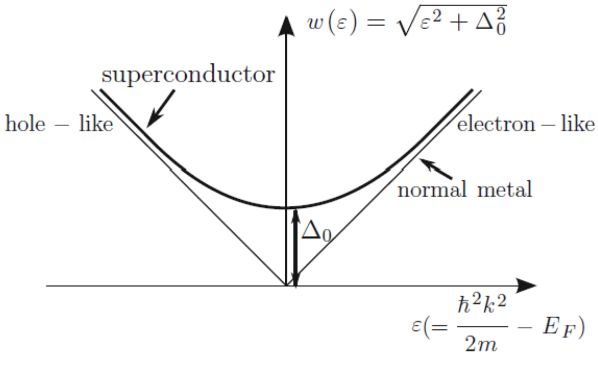


Figure 8: Comparison of the elementary excitation spectra for a normal metal and a superconductor at $T = 0$. The superconducting spectrum $\omega(\xi) = \sqrt{\xi^2 + \Delta_0^2}$ displays a clear energy gap Δ_0 at the Fermi level ($\xi = 0$).

The energy gap Δ_0 fundamentally alters the available states for excitations. The density of quasiparticle states $D_S(\omega)$ in the superconductor is derived from the dispersion relation defined in Eq. 82. Given that constant energy surfaces in reciprocal space remain spherical near the Fermi surface, the number of states in the interval $(\omega, \omega + d\omega)$ is:

$$D_S(\omega)d\omega = \frac{2V}{(2\pi)^3}4\pi k^2 dk \simeq \frac{Vk_F^2}{\pi^2}dk \tag{84}$$

where k is approximated by k_F for energies sufficiently close to the Fermi level. Differentiating the dispersion relation $\omega = \sqrt{\xi^2 + \Delta_0^2}$ with respect to k yields:

$$\frac{d\omega}{dk} = \frac{\xi}{\sqrt{\xi^2 + \Delta_0^2}}\frac{d\xi}{dk} = \frac{\sqrt{\omega^2 - \Delta_0^2}}{\omega}\frac{\hbar^2 k_F}{m} \tag{85}$$

By substituting this derivative into the expression for $D_S(\omega)$, we obtain the density of states for the superconductor:

$$D_S(\omega) = D(E_F)\frac{\omega}{\sqrt{\omega^2 - \Delta_0^2}} \tag{86}$$

where $D(E_F) = \frac{mk_F}{\pi^2\hbar^2}V$ is the density of states of the normal metal at the Fermi energy. To represent this on an absolute energy scale E, the expression is recast as:

$$D_S(E) = D(E_F)\frac{|E - E_F|}{\sqrt{(E - E_F)^2 - \Delta_0^2}} \quad \text{for } |E - E_F| > \Delta_0 \tag{87}$$

This result implies a divergence in the density of states at the gap edges $E = E_F \pm \Delta_0$, known as the coherence peaks or **Van Hove singularities**. As shown in Figure 9, the states that are excluded from the gap region $[E_F - \Delta_0, E_F + \Delta_0]$ are redistributed immediately outside the gap. Furthermore, since breaking a single Cooper pair results in the creation of two quasiparticles, the minimum energy required for such a process is $2\Delta_0$. This value represents the collective binding energy of a pair within the superconducting condensate.

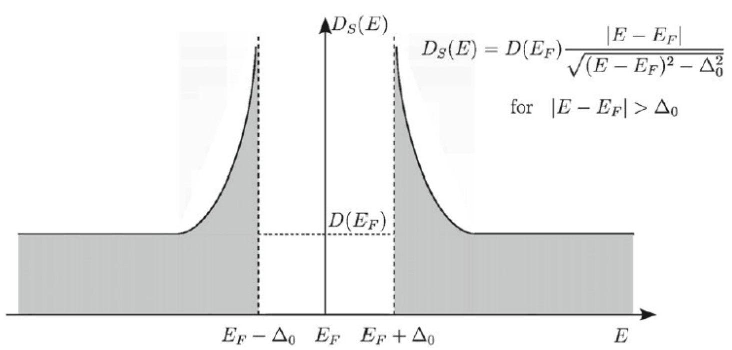


Figure 9: Density of states for a superconductor. The superconducting DOS exhibits a gap of width $2\Delta_0$ centered at E_F and characteristic singularities at the gap edges.

4.2 Persistent currents.

The existence of an energy gap Δ_0 at the Fermi level fundamentally alters the transport properties of the superconducting state. These modifications affect phenomena such as perfect conductance, diamagnetic response, tunneling, specific heat, and ultrasonic attenuation. To understand the zero electrical resistance (persistent currents), we must extend the BCS ground state to a current-carrying state.

In the quiescent ground state, electrons are paired in states $(\vec{k}\uparrow, -\vec{k}\downarrow)$ with a total momentum $\vec{K} = \vec{k} - \vec{k} = 0$. A current-carrying state is constructed by pairing electrons with a common center-of-mass momentum, represented by the states $(\vec{k} + \vec{Q}\uparrow, -\vec{k} + \vec{Q}\downarrow)$. The total momentum for each Cooper pair is $\hbar\vec{K} = \hbar(\vec{k} + \vec{Q} - \vec{k} + \vec{Q}) = 2\hbar\vec{Q}$. The associated kinetic energy of the center-of-mass motion for each pair is:

$$E_{kin} = \frac{\hbar^2(2Q)^2}{2(2m)} \tag{88}$$

For values of $\vec{Q}$ typically encountered in transport, E_{kin} is significantly smaller than the binding energy $2\Delta_0$. Consequently, **the kinetic energy is insufficient to break individual pairs**.

Degrading the current via standard scattering mechanisms would require changing the momentum of a Cooper pair relative to the collective value $2\vec{Q}$. Due to the cooperative nature of the superconducting condensate, the energy cost to displace a single pair from the common momentum state is approximately the binding energy $2\Delta_0$. As the number of pairs with randomized momenta increases, the energy penalty becomes prohibitively large. Furthermore, astray pairs naturally tend to "condense" back into the coherent state where all pairs share the same momentum to maximize the energy gain. A current-degrading process would necessitate the simultaneous scattering of a macroscopic number of pairs into a new coherent state with a different total momentum—a highly improbable event. This rigidity of the wavefunction justifies the existence of persistent currents and the absence of electrical resistance.

5. Treatment of superconductors at finite temperature and heat capacity.

Self-consistency at finite temperature.

At finite temperature, the quasiparticle states within the superconductor become thermally excited, leading to the breaking of Cooper pairs. This process results in a progressive reduction of the energy gap, eventually driving the system into the normal state at the critical temperature T_c.

To describe this behavior, we extend the mean-field approach by considering the thermal average of the pair operators. The product $c^{\dagger}_{\vec{k}\uparrow}c^{\dagger}_{-\vec{k}\downarrow}$ is expressed in terms of its thermal expectation value $a_{\vec{k}}$ and its corresponding fluctuation operator:

$$c^{\dagger}_{\vec{k}\uparrow}c^{\dagger}_{-\vec{k}\downarrow} = a_{\vec{k}} + (c^{\dagger}_{\vec{k}\uparrow}c^{\dagger}_{-\vec{k}\downarrow} - a_{\vec{k}}) \tag{89}$$

where $a_{\vec{k}}$ is now defined as the thermal average:

$$a_{\vec{k}} = \langle c^{\dagger}_{\vec{k}\uparrow}c^{\dagger}_{-\vec{k}\downarrow}\rangle_T \tag{90}$$

Following the Bogoliubov procedure, the temperature-dependent gap parameter $\Delta_{\vec{k}}(T)$ is defined by the generalization of the self-consistency condition:

$$\Delta_{\vec{k}} = -\sum_{\vec{k}'} V_{\vec{k}\vec{k}'}\langle c_{-\vec{k}'\downarrow}c_{\vec{k}'\uparrow}\rangle_T \tag{91}$$

By applying the inverse Bogoliubov transformations, we express the operator product in terms of the quasiparticle operators γ. The thermal average then becomes:

$$\begin{aligned}\langle c_{-\vec{k}\downarrow}c_{\vec{k}\uparrow}\rangle_T &= \langle (u_{\vec{k}}\gamma_{-\vec{k}\downarrow} - v_{\vec{k}}\gamma^{\dagger}_{\vec{k}\uparrow})(u_{\vec{k}}\gamma_{\vec{k}\uparrow} + v_{\vec{k}}\gamma^{\dagger}_{-\vec{k}\downarrow})\rangle_T = \\ &= u_{\vec{k}}v_{\vec{k}}\langle 1 - \gamma^{\dagger}_{-\vec{k}\downarrow}\gamma_{-\vec{k}\downarrow} - \gamma^{\dagger}_{\vec{k}\uparrow}\gamma_{\vec{k}\uparrow}\rangle_T = u_{\vec{k}}v_{\vec{k}}(1 - 2f(\omega_{\vec{k}}))\end{aligned} \tag{92}$$

where $f(E) = (e^{\beta E} + 1)^{-1}$ is the Fermi-Dirac distribution function and $\omega_{\vec{k}} = \sqrt{\xi^2_{\vec{k}} + \Delta^2_{\vec{k}}}$ represents the quasiparticle energies. Using the identity $1 - 2f(E) = \tanh\frac{\beta E}{2}$, the self-consistent gap equation at finite temperature takes the form:

$$\Delta_{\vec{k}} = -\frac{1}{2}\sum_{\vec{k}'} V_{\vec{k}\vec{k}'}\frac{\Delta_{\vec{k}'}}{\sqrt{\xi^2_{\vec{k}'} + \Delta^2_{\vec{k}'}}}\tanh\left(\frac{\beta\sqrt{\xi^2_{\vec{k}'} + \Delta^2_{\vec{k}'}}}{2}\right) \tag{93}$$

This integral equation determines the evolution of the gap $\Delta(T)$ from its maximum value at zero temperature until it vanishes at T_c, where the thermal energy is sufficient to completely overcome the pairing interaction.

In the average potential approximation, the self-consistent gap equation simplifies into an **implicit relation for the temperature dependence of the gap parameter** $\Delta(T)$:

$$1 = \frac{1}{2}U_0 n_0 \int_{-\hbar\omega_D}^{\hbar\omega_D}\frac{d\xi}{\sqrt{\xi^2 + \Delta^2}}\tanh\left(\frac{\beta\sqrt{\xi^2 + \Delta^2}}{2}\right) \tag{94}$$

where n_0 denotes the density-of-states for one spin direction per unit cell at the Fermi energy. As the temperature T increases, the hyperbolic tangent term decreases. To maintain the equality, the energy gap Δ must cooperatively decrease, eventually vanishing at a critical temperature T_c. This transition to the normal state occurs when

$\Delta(T_c) = 0$, leading to the following condition:

$$1 = U_0 n_0 \int_0^{\hbar\omega_D} \frac{1}{\xi} \tanh\left(\frac{\xi}{2k_B T_c}\right) d\xi \tag{95}$$

By introducing the dimensionless variable $x = \frac{\xi}{2k_B T_c}$, we can evaluate the integral in the limit where $a = \frac{\hbar\omega_D}{2k_B T_c} \gg 1$. Using the property:

$$\int_0^a \frac{\tanh x}{x} dx \simeq \ln\left(\frac{4\gamma a}{\pi}\right) \tag{96}$$

where $\gamma \simeq 1.781\ldots$ is the Euler constant, we obtain:

$$\ln\left(\frac{1.13\hbar\omega_D}{k_B T_c}\right) = \frac{1}{U_0 n_0} \Leftrightarrow k_B T_c = 1.13\hbar\omega_D e^{-\frac{1}{U_0 n_0}} \tag{97}$$

Comparing this result with the zero-temperature gap $\Delta(0) =: \Delta_0$ derived in Eq. 49, we find a universal BCS ratio:

$$\Delta(0) = 1.76 k_B T_c \tag{98}$$

The numerical solution for $\Delta(T)$ shows that near the critical temperature ($T \to T_c$), the gap vanishes following a power law:

$$\Delta(T) = 3.06 k_B T_c \left(1 - \frac{T}{T_c}\right)^{\frac{1}{2}} \tag{99}$$

The exponent $\frac{1}{2}$ is a characteristic signature of mean-field theory transitions. The overall temperature dependence is illustrated in Figure 10.

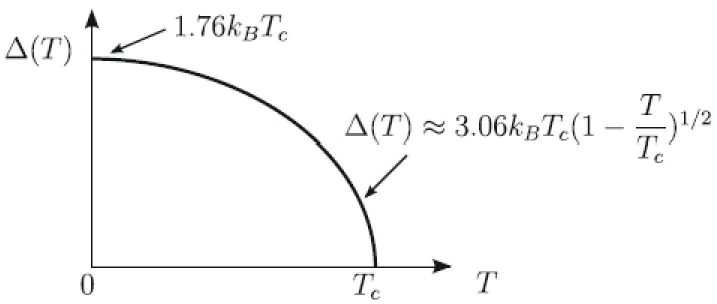


Figure 10: Behavior of the energy gap parameter $\Delta(T)$ for a superconductor in the BCS theory within the weak coupling limit. The gap remains relatively constant at low temperatures and drops sharply to zero as T approaches T_c.

Electronic heat capacity for a superconductor.

The electronic heat capacity of a superconductor undergoes a transformative change at the transition temperature, as illustrated in Figure 11. This transition is marked by a sharp discontinuity at the critical temperature T_c. For $T > T_c$, the electronic heat capacity of the normal material remains linear with temperature. However, for $T \to 0$, **the heat capacity decays exponentially**, a behavior consistent with the existence of an energy gap in the quasiparticle excitation spectrum.

To calculate the electronic heat capacity, we begin with the entropy $S(T)$ for a system of fermionic quasiparticles:

$$S(T) = -2k_B \sum_{\vec{k}} \left(f_{\vec{k}} \ln f_{\vec{k}} + (1 - f_{\vec{k}}) \ln(1 - f_{\vec{k}}) \right) \tag{100}$$

where the factor 2 accounts for spin orientations and $f_{\vec{k}}$ is the Fermi-Dirac distribution:

$$f_{\vec{k}} = \left(e^{\beta \omega_{\vec{k}}} + 1 \right)^{-1} \quad \text{with} \quad \omega_{\vec{k}} = \sqrt{\xi_{\vec{k}}^2 + \Delta^2} \quad \text{and} \quad \Delta := \Delta(T) \tag{101}$$

The heat capacity is defined as $C(T) = T\frac{dS(T)}{dT} = \sum_{\vec{k}} \frac{\partial S}{\partial f_{\vec{k}}} \frac{\partial f_{\vec{k}}}{\partial T}$. Evaluating the first partial derivative, we find:

$$\frac{\partial S}{\partial f_{\vec{k}}} = -2k_B \ln \left(\frac{f_{\vec{k}}}{1 - f_{\vec{k}}} \right) = \frac{2\omega_{\vec{k}}}{T} = \frac{2}{T} \sqrt{\xi_{\vec{k}}^2 + \Delta^2} \tag{102}$$

The derivative of the distribution function with respect to temperature is given by:

$$\begin{aligned} \frac{\partial f_{\vec{k}}}{\partial T} =& \frac{1}{k_B T^2} \frac{e^{\beta \omega_{\vec{k}}}}{\left(e^{\beta \omega_{\vec{k}}} + 1 \right)^2} \left(\omega_{\vec{k}} - T \frac{d\omega_{\vec{k}}}{dT} \right) = \\ =& \frac{1}{k_B T^2} \frac{e^{\beta \omega_{\vec{k}}}}{\left(e^{\beta \omega_{\vec{k}}} + 1 \right)^2} \left(\sqrt{\xi_{\vec{k}}^2 + \Delta^2} - \frac{T}{2} \frac{1}{\sqrt{\xi_{\vec{k}}^2 + \Delta^2}} \frac{d\Delta^2}{dT} \right) \end{aligned} \tag{103}$$

Combining these terms into the sum for $C(T)$, we obtain:

$$C(T) = \frac{2}{k_B T^2} \sum_{\vec{k}} \frac{e^{\beta \omega_{\vec{k}}}}{\left(e^{\beta \omega_{\vec{k}}} + 1 \right)^2} \left(\xi_{\vec{k}}^2 + \Delta^2 - \frac{T}{2} \frac{d\Delta^2}{dT} \right) \tag{104}$$

By converting the summation over $\vec{k}$ into an integral over energy ξ with the density of states $D(E_F)$ for both spin directions, we arrive at:

$$C(T) = \frac{D(E_F)}{k_B T^2} \int_{-\infty}^{\infty} \frac{e^{\beta \sqrt{\xi^2 + \Delta^2}}}{\left(e^{\beta \sqrt{\xi^2 + \Delta^2}} + 1 \right)^2} \left(\xi^2 + \Delta^2 - \frac{T}{2} \frac{d\Delta^2}{dT} \right) d\xi \tag{105}$$

In the normal state ($\Delta = 0$), the integral simplifies to $C_n = \frac{\pi^2}{3} D(E_F) k_B^2 T$. At $T = T_c$, a discontinuity occurs due to the term involving $\frac{d\Delta^2}{dT}$, which equals $-9.36 k_B^2 T_c$ in the BCS model. The magnitude of this jump is:

$$\Delta C(T_c) \simeq 9.36 k_B D(E_F) \int_0^\infty \frac{e^{\beta\xi}}{(e^{\beta\xi}+1)^2} d\xi \simeq 4.68 D(E_F) k_B^2 T_c \tag{106}$$

This leads to the parameter-free BCS ratio for the specific heat jump:

$$\frac{C_s - C_n}{C_n} \simeq \frac{4.68 \cdot 3}{\pi^2} \simeq 1.42 \tag{107}$$

For $T \ll T_c$, the heat capacity is dominated by the energy gap, exhibiting an exponential decay proportional to $e^{-\frac{\Delta(0)}{k_B T}} = e^{-1.76\frac{T_c}{T}}$.

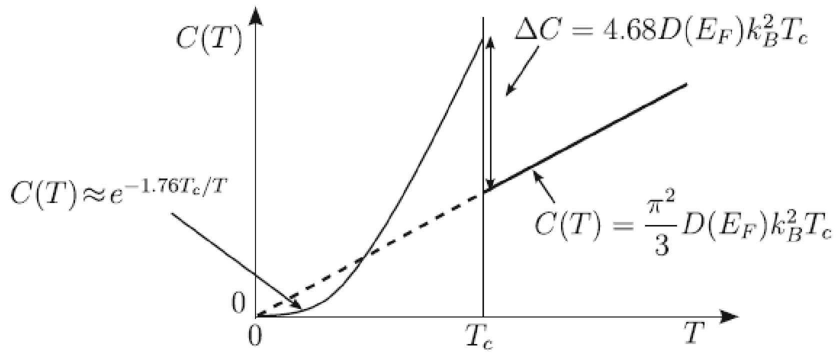


Figure 11: Electronic heat capacity of a normal metal and a superconductor. Note the linear behavior for $T > T_c$, the jump at T_c, and the exponential suppression as $T \to 0$.

6. The phenomenological London model.

Diamagnetism and Meissner effect.

The phenomenological description of superconductors in the presence of magnetic fields involves distinctive behaviors previously introduced, most notably the Meissner effect, which dictates that a weak magnetic field is incapable of penetrating the bulk of a superconducting material. To provide a theoretical foundation for this phenomenon, F. and H. London proposed a model in 1935 that **elucidates the transport and magnetic properties distinguishing ordinary conductors, perfect conductors, and superconductors**. In a standard conductor characterized by an electron density n and effective mass m, the charge carriers undergo scattering processes due to impurities, phonons, and lattice defects. By assuming a simplified carrier dynamics governed by a finite relaxation time τ, the average drift velocity is expressed as $\vec{v}_d = -e\tau\vec{E}$, leading to the classical Ohm's law $\vec{J} = \sigma\vec{E}$, where the conductivity is defined as $\sigma = \frac{ne^2\tau}{m}$.

In contrast, we consider an ideally pure metal featuring a parabolic conduction band and a density n of electrons with effective mass m. Under the influence of an external electric field $\vec{E}$, the motion of these freely moving electrons is governed by the ballistic dynamic equation $m\frac{d\vec{v}}{dt} = -e\vec{E}$. Consequently, the relationship between the current density and the electric field is established as:

$$\vec{J} = n(-e)\vec{v} \Rightarrow \frac{\partial \vec{J}}{\partial t} = \frac{ne^2}{m}\vec{E} \tag{108}$$

This expression constitutes the first London equation, derived under the assumption of ideal collisionless motion. It is noteworthy that for a standard conductor with a finite τ, the term on the left-hand side would be replaced by $\frac{\vec{J}}{\tau}$; thus, the first London equation represents a fundamental modification of Ohm's law to describe the acceleration of collisionless electrons. By applying the curl operator to both sides of this equation and utilizing the Maxwell equation for the electric field, we derive:

$$\frac{\partial}{\partial t}\left(\vec{\nabla} \times \vec{J}\right) = \frac{ne^2}{m}\vec{\nabla} \times \vec{E} = -\frac{ne^2}{mc}\frac{\partial \vec{B}}{\partial t} \Rightarrow \frac{\partial}{\partial t}\left(\vec{\nabla} \times \vec{J} + \frac{ne^2}{mc}\vec{B}\right) = 0 \tag{109}$$

While this result applies to any ideal conductor, it is insufficient to explain the Meissner effect on its own. The equation indicates that the vector field defined by the quantity within the parentheses is constant over time, yet it does not specify whether fields and currents are excluded from the sample interior.

The macroscopic London theory postulates that the electronic density in the superconducting state consists of normal electrons and super-electrons, the latter condensing into a macroscopic quantum state. The London brothers conjectured that if the term in parentheses for super-electrons is not merely time-independent but vanishes identically, the ideal conductor will exhibit the Meissner effect. This leads to the second London equation:

$$\vec{\nabla} \times \vec{J}_s = -\frac{n_s e^2}{mc}\vec{B} \tag{110}$$

This fundamental relation serves as the basis for the exclusion of magnetic fields and currents from the interior of superconducting materials.

The first London equation, as expressed in Eq. 108, implies that a perfect conductor under stationary conditions cannot sustain an internal electric field. Similarly, the second London equation, shown in Eq. 110, establishes that a superconductor in a stationary state cannot maintain a magnetic field in its interior, with the exception of a thin surface layer. The characteristic penetration depth is derived by relating the magnetic field to the current density through the Maxwell equation:

$$\vec{\nabla} \times \vec{B} = \frac{4\pi}{c}\vec{J}_s \tag{111}$$

By applying the curl operator to both sides of the previous expression and substituting the second London relation, we obtain:

$$\vec{\nabla} \times \left(\vec{\nabla} \times \vec{B}\right) = \frac{4\pi}{c} \vec{\nabla} \times \vec{J}_s = -\frac{4\pi n_s e^2}{mc^2} \vec{B} \tag{112}$$

Utilizing the vector identity $\vec{\nabla} \times \left(\vec{\nabla} \times \vec{A}\right) = \vec{\nabla}\left(\vec{\nabla} \cdot \vec{A}\right) - \nabla^2 \vec{A}$ and the Maxwell condition $\vec{\nabla} \cdot \vec{B} = 0$, we arrive at:

$$\nabla^2 \vec{B}\left(\vec{r}\right) = \frac{1}{\lambda_L^2} \vec{B}\left(\vec{r}\right) \tag{113}$$

where the London penetration length λ_L is defined as:

$$\lambda_L = \frac{c}{\omega_p} = \sqrt{\frac{mc^2}{4\pi n_s e^2}} \tag{114}$$

In this definition, ω_p denotes the plasma frequency corresponding to the super-electron density n_s. For standard metallic electron concentrations, and assuming ballistic behavior for all electrons ($n = n_s$), the value of λ_L is typically on the order of 10^2 to 10^3 Å. While it is often assumed at $T = 0$ that all electrons behave as super-electrons, a more sophisticated theory is necessary to describe the temperature dependence of n_s.

To examine the implications of these equations, we consider a semi-infinite superconducting slab with its surface in the xy plane, where the region $z < 0$ is vacuum. Two specific configurations are particularly illustrative. First, consider a magnetic field $\vec{B}$ parallel to the z-axis and homogeneous in the xy plane, given by $\vec{B} = (0, 0, B(z))$. The condition $\vec{\nabla} \cdot \vec{B} = 0$ implies that $\frac{\partial B(z)}{\partial z} = 0$, requiring the field to be constant. However, the governing Helmholtz equation forces $B(z) = 0$ for $z > 0$, demonstrating that a magnetic field cannot exist normal to the superconductor surface.

Second, consider a magnetic field $\vec{B}$ parallel to the x-axis and homogeneous in the xy plane, as shown in Figure 12.

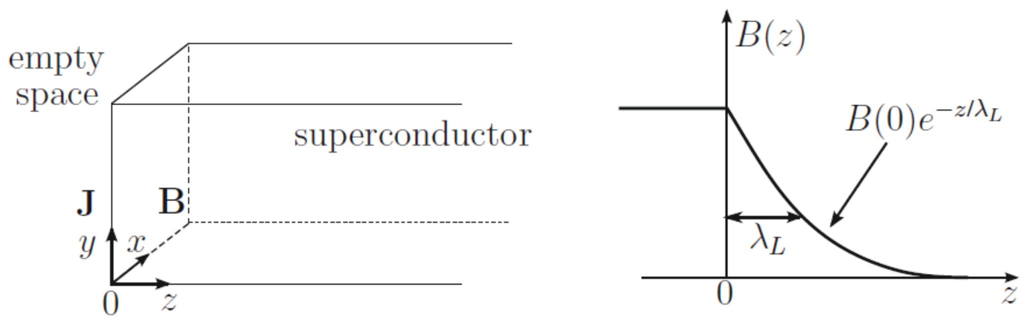


Figure 12: Semi-infinite slab geometry illustrating the screening currents $\vec{J}_s$ and the exponential decay of the magnetic field $\vec{B}(z)$ parallel to the surface within the London penetration depth λ_L.

In this case, $\vec{B} = (B(z), 0, 0)$ satisfies the divergence condition automatically. The field equation reduces to:

$$\frac{\partial^2 B(z)}{\partial z^2} = \frac{1}{\lambda_L^2} B(z) \tag{115}$$

The solution for the region $z > 0$ is expressed as:

$$B(z) = B(0)\, e^{-\frac{z}{\lambda_L}} \tag{116}$$

This indicates that, **beyond a narrow surface layer defined by** λ_L, the magnetic field parallel to the surface cannot penetrate the bulk of the superconductor, which is a clear manifestation of the Meissner effect. An analogous behavior is observed for the screening currents $\vec{J}_s$ directed along the y-axis in this geometry.

The London gauge for superconductors.

To conclude this section, it is essential to examine the further implications of the second London equation under stationary conditions. In this regime, the continuity equation for the current density ensures that its divergence vanishes. For the following derivation, we assume a superconducting sample of finite volume and simply connected geometry. The second London equation can be expressed in a more mathematically convenient form by introducing the vector potential $\vec{A}(\vec{r})$, defined through the magnetic induction as:

$$\vec{B}(\vec{r}) = \vec{\nabla} \times \vec{A}(\vec{r}) \tag{117}$$

As is standard in electromagnetism, this definition allows for an arbitrary gauge transformation of the form $\vec{A}'(\vec{r}) = \vec{A}(\vec{r}) + \vec{\nabla}\phi(\vec{r})$, where $\phi(\vec{r})$ is a regular, single-valued scalar function. Substituting the curl of the vector potential into the second London equation yields:

$$\vec{\nabla} \times \left(\vec{J}_s(\vec{r}) + \frac{n_s e^2}{mc} \vec{A}(\vec{r}) \right) = 0 \tag{118}$$

The expression within the parentheses is irrotational inside the superconductor and can therefore be represented as the gradient of a scalar field $\chi(\vec{r})$. Given the simply connected nature of the geometry, we can write $\vec{J}_s(\vec{r}) + \frac{n_s e^2}{mc}\vec{A}(\vec{r}) = -\vec{\nabla}\chi(\vec{r})$. By exploiting the gauge freedom inherent in $\vec{A}(\vec{r})$, we can absorb the term $\vec{\nabla}\chi(\vec{r})$ into the vector potential itself, leading to the local London equation:

$$\vec{J}_s(\vec{r}) = -\frac{n_s e^2}{mc} \vec{A}(\vec{r}) \tag{119}$$

This relation demonstrates that for connected superconductors, a specific gauge exists—known as the London gauge—where the supercurrent density and the vector potential are directly proportional. This equation is not gauge-invariant and is uniquely defined by the following set of conditions:

$$\begin{cases} \vec{\nabla} \times \vec{A}(\vec{r}) = \vec{B}(\vec{r}) & \text{Fundamental relation} \\ \vec{\nabla} \cdot \vec{A}(\vec{r}) = 0 & \text{Bulk condition} \\ \vec{A}(\vec{r}_s) \cdot \hat{n} = -\frac{mc}{n_s e^2} \vec{J}_s(\vec{r}_s) \cdot \hat{n} & \text{Surface condition} \end{cases} \tag{120}$$

In this gauge, the Coulomb condition $\vec{\nabla} \cdot \vec{A} = 0$ is consistent with the stationary requirement $\vec{\nabla} \cdot \vec{J}_s = 0$. Furthermore, the normal component of the vector potential at the surface is determined by the normal component of the current density; if no external current is injected, both vanish at the boundary.

The electrodynamics described by the London model is strictly local, meaning the current density at a specific coordinate depends solely on the vector potential at that same point. This framework holds when the vector potential varies slowly over the scale of the BCS coherence length ξ_0. While **this local assumption is highly accurate for type-II superconductors and type-I materials near the transition temperature**, it fails when non-local effects become dominant. In such instances, a non-local electrodynamics, as proposed by Pippard, is required. This approach recognizes that the current density is actually a weighted integral of the vector potential over a volume characterized by the correlation length of the Cooper pairs and the mean free path of the electrons.

7. Macroscopic quantum phenomena.

7.1 Order parameter in superconductors and the Ginzburg-Landau equations.

Moving beyond the local London model, the Ginzburg-Landau (GL) theory provides a powerful phenomenological framework for understanding inhomogeneous superconductors, boundary effects, and the **transition between normal and superconducting states**. This approach sidesteps the technical complexity of many-body wavefunctions by introducing a macroscopic complex order parameter, $\psi(\vec{r})$, which characterizes the degree of superconducting order. Below the critical temperature T_c, the local density of Cooper pairs is directly related to this parameter by $n_{\text{pairs}}(\vec{r}) = |\psi(\vec{r})|^2$.

The fundamental postulate of the GL theory is that the free energy density f_S near the transition temperature can be expanded in terms of the order parameter and the vector potential $\vec{A}$:

$$f_S(T, \psi, \vec{A}) = f_N(T) + \alpha(T)|\psi|^2 + \frac{1}{2}\beta(T)|\psi|^4 + \frac{1}{2m^*} \left| \left(\vec{p} - \frac{e^*}{c}\vec{A} \right) \psi(\vec{r}) \right|^2 + \frac{\vec{B}^2}{8\pi} \tag{121}$$

In this expression, $f_N(T)$ represents the free energy density of the normal phase where $\psi = 0$. The total free energy of the superconducting volume V is then determined by the volume integral:

$$F_S(T) = \int_V f_S(T, \psi(\vec{r}), \vec{A}(\vec{r}))d\vec{r} \tag{122}$$

The expansion in powers of $|\psi|^2$ and $|\psi|^4$ is justified near the critical temperature, where **the order parameter is sufficiently small to allow the omission of higher-order terms**. The fourth term in the free energy density is constructed by treating $\psi(\vec{r})$ as a macroscopic quantum wavefunction for a particle with effective charge e^* and mass m^* in the presence of spatial gradients and magnetic fields. In alignment with the BCS microscopic theory, these parameters are identified as $e^* = -2e$ and $m^* = 2m$, reflecting the nature of the Cooper pair as the fundamental superconducting carrier. Finally, the last term corresponds to the electromagnetic energy density associated with the magnetic field $\vec{B} = \vec{\nabla} \times \vec{A}$.

The spatial distribution of the order parameter $\psi(\vec{r})$ is determined by the minimization of the total free energy $F_S(T)$ with respect to arbitrary variations of its complex conjugate ψ^*. This variational procedure yields the celebrated Ginzburg-Landau equation:

$$\frac{1}{2m^*}\left(\vec{p} - \frac{e^*}{c}\vec{A}\right)^2 \psi(\vec{r}) + \beta|\psi(\vec{r})|^2\psi(\vec{r}) = -\alpha(T)\psi(\vec{r}) \tag{123}$$

This expression is inherently nonlinear due to the cubic term in ψ, reflecting the interactions within the superconducting condensate. Complementing this, the supercurrent density $\vec{J}_s(\vec{r})$ is derived in a form analogous to the probability current in standard quantum mechanics:

$$\vec{J}_s(\vec{r}) = \frac{-i\hbar e^*}{2m^*}\left(\psi^*(\vec{r})\vec{\nabla}\psi(\vec{r}) - \psi(\vec{r})\vec{\nabla}\psi^*(\vec{r})\right) - \frac{e^{*2}}{m^*c}|\psi(\vec{r})|^2\vec{A}(\vec{r}) \tag{124}$$

The current density equation can also be obtained through the minimization of $F_S(T)$ with respect to variations of the vector potential $\vec{A}$. By adopting a polar representation for the macroscopic wavefunction, $\psi(\vec{r}) = |\psi(\vec{r})|e^{i\theta(\vec{r})}$, where $\theta(\vec{r})$ is the local phase, the expression for the current density simplifies significantly:

$$\vec{J}_s(\vec{r}) = |\psi(\vec{r})|^2\left(\frac{e^*\hbar}{m^*}\vec{\nabla}\theta(\vec{r}) - \frac{e^{*2}}{m^*c}\vec{A}(\vec{r})\right) \tag{125}$$

Physical intuition suggests that $\psi(\vec{r})$ **describes the collective motion of the center of mass of the Cooper pair condensate**. The squared modulus $|\psi(\vec{r})|^2$ corresponds to the local density of pairs, while the gradient of the phase $\vec{\nabla}\theta(\vec{r})$ governs the local velocity of the center of mass in the limit of vanishing magnetic fields.

The variational derivation of these equations involves considering an infinitesimal variation $\psi^* \to \psi^* + \delta\psi^*$. The resulting change in the total free energy is given by:

$$\delta F_S = \int_V \left(\alpha\delta|\psi|^2 + \frac{1}{2}\beta\delta|\psi|^4 + \frac{1}{2m^*}\delta \left| \left(\vec{p} - \frac{e^*}{c}\vec{A} \right) \psi \right|^2 \right) d\vec{r} \tag{126}$$

The first two terms of the integrand are straightforwardly expanded as $\delta\psi^*(\alpha\psi + \beta|\psi|^2\psi)$. The kinetic energy term requires integration by parts; by applying Green's theorem and neglecting surface integrals under appropriate boundary conditions, we obtain:

$$\int \left(i\hbar\vec{\nabla} + \frac{e^*}{c}\vec{A} \right) \delta\psi^* \cdot \left(-i\hbar\vec{\nabla} - \frac{e^*}{c}\vec{A} \right) \psi d\vec{r} = \int \delta\psi^* \left(-i\hbar\vec{\nabla} - \frac{e^*}{c}\vec{A} \right)^2 \psi d\vec{r} \tag{127}$$

Setting the total variation $\delta F_S = 0$ for an arbitrary $\delta\psi^*$ leads directly to the Ginzburg-Landau equation. A similar minimization with respect to the vector potential, incorporating the Maxwell relation $\vec{J} = \frac{c}{4\pi}\vec{\nabla} \times \vec{B}$, recovers the current density expression.

7.2 The phenomenological parameters of the Ginzburg-Landau equations.

The free energy density of a superconductor is characterized by two phenomenological parameters, $\alpha(T)$ and $\beta(T)$. To elucidate their physical significance, we first examine a bulk superconductor in the **absence of external magnetic fields and spatial gradients** ($\vec{A}(\vec{r}) = \vec{\nabla}\psi(\vec{r}) = 0$). Under these conditions, the difference in free energy density between the superconducting and normal phases simplifies to:

$$f_S(T, \psi) - f_N(T) = \alpha(T)|\psi|^2 + \frac{1}{2}\beta(T)|\psi|^4 \tag{128}$$

This expression represents a Taylor expansion of the free energy change in terms of the order parameter magnitude. For the superconducting state to be thermodynamically stable, these parameters must satisfy specific constraints: $\beta(T)$ must remain positive, while $\alpha(T)$ must be positive for $T > T_c$ and negative for $T < T_c$ (see Fig. 13). Such conditions ensure that the free energy possesses a global minimum at a non-zero value of $|\psi|$ when the temperature drops below the critical threshold.

The simplest linear approximation for these parameters in the vicinity of T_c is given by:

$$\begin{cases} \alpha(T) = \alpha_0(T - T_c) & \text{with } \alpha_0 > 0 \\ \beta(T) = \beta_0 & \text{with } \beta_0 > 0 \end{cases} \tag{129}$$

In this framework, we assume $\alpha(T)$ varies linearly with the temperature deviation from the critical point, while the temperature dependence of β is considered negligible for $T \sim T_c$. By further analyzing the structure of the Ginzburg-Landau equations, it **becomes possible to relate these phenomenological constants to experimentally accessible macroscopic quantities**, including the thermodynamic critical field $H_c(T)$ and the effective magnetic penetration depth $\lambda_{\text{eff}}(T)$, which are standard parameters for characterizing isotropic superconducting systems.

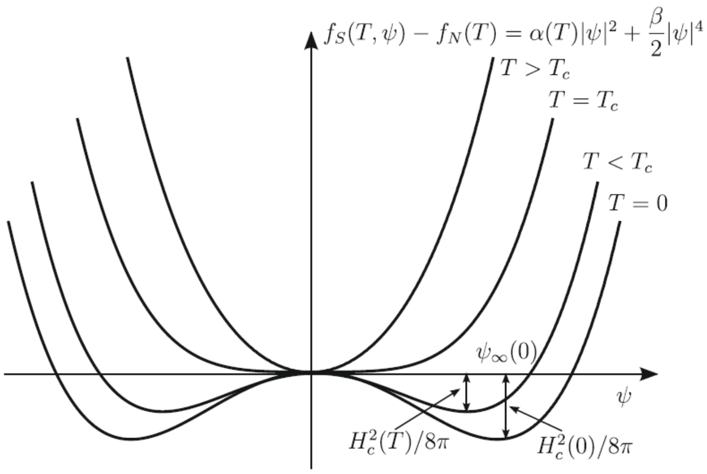


Figure 13: Temperature dependence of the Ginzburg-Landau free energy density as a function of the order parameter, showing the transition from a single minimum at $\psi = 0$ to a degenerate set of minima at finite $|\psi|$.

Thermodynamic critical field.

By analyzing Eq. 123 for a bulk superconductor in the limit where external fields and spatial gradients are absent ($\vec{A}(\vec{r}) = 0$ and $\psi(\vec{r}) = \text{const} = \psi_\infty$), the Ginzburg-Landau equation simplifies significantly. For temperatures below T_c, the relation $\beta|\psi(\vec{r})|^2 = -\alpha(T)$ must hold, which determines the equilibrium value of the order parameter:

$$|\psi_\infty|^2 = -\frac{\alpha(T)}{\beta} \quad \text{for } T < T_c \tag{130}$$

Conversely, for temperatures above T_c, the only stable solution is $\psi_\infty = 0$. By substituting this equilibrium density into the free energy density expansion, the difference in free energy between the superconducting and normal phases is expressed as:

$$f_S(T, \psi_\infty) - f_N(T) = -\frac{\alpha^2(T)}{2\beta} \quad \text{for } T < T_c \tag{131}$$

This term represents the condensation energy per unit volume, which can be identified with the magnetic energy density $-\frac{H_c^2(T)}{8\pi}$. Using the empirical value of the thermodynamic critical field $H_c(T)$, we establish a fundamental relationship between the phenomenological parameters and the critical field:

$$\frac{H_c^2(T)}{8\pi} = \frac{\alpha^2(T)}{2\beta} \Rightarrow H_c(T) = 2\sqrt{\frac{\pi}{\beta}}|\alpha(T)| \tag{132}$$

Given the temperature dependence of $\alpha(T)$ defined previously, it follows that $H_c(T)$ vanishes linearly as $(T_c - T)$ when the system approaches the critical temperature from below.

Magnetic penetration depth.

Assuming that the order parameter $\psi(\vec{r})$ remains constant and equal to its bulk value ψ_∞ up to the material interface, we can analyze the magnetic penetration depth within the Ginzburg-Landau framework. Under this assumption, the supercurrent density expression from Eq. 124 simplifies significantly. By substituting the supercurrent into the relevant Maxwell equation, $\vec{\nabla} \times \vec{B} = \frac{4\pi}{c}\vec{J}_s$, we obtain:

$$\vec{J}_s(\vec{r}) = -\frac{e^{*2}}{m^* c}|\psi_\infty|^2 \vec{A}(\vec{r}) \Rightarrow \frac{c}{4\pi}\vec{\nabla} \times \vec{B} = -\frac{e^{*2}}{m^* c}|\psi_\infty|^2 \vec{A}(\vec{r}) \tag{133}$$

Applying the curl operator to both sides of this relation allows us to identify the temperature-dependent effective penetration depth $\lambda_{\text{eff}}(T)$. The resulting Helmholtz equation for the magnetic field yields the following expression for the penetration length:

$$\frac{1}{\lambda_{\text{eff}}^2(T)} = \frac{4\pi e^{*2}}{m^* c^2}|\psi_\infty|^2 \Rightarrow \lambda_{\text{eff}}^2(T) = \frac{m^* c^2 \beta}{4\pi e^{*2}|\alpha(T)|} \tag{134}$$

Based on the previously established linear temperature dependence of $\alpha(T)$, it is evident that $\lambda_{\text{eff}}(T)$ exhibits a divergence proportional to $(T_c - T)^{-\frac{1}{2}}$ as the system approaches the critical temperature from below. By algebraically combining the expressions for the thermodynamic critical field H_c and this effective penetration depth λ_{eff}, the phenomenological parameters α and β **can be uniquely determined as functions of experimentally measurable quantities**.

7.2.1 Ginzburg-Landau coherence length.

Beyond the magnetic penetration depth, the Ginzburg-Landau equations introduce a second characteristic length scale that governs the spatial variations of the order parameter. To illustrate this, we consider a semi-infinite superconductor occupying the half-plane $z > 0$, where the order parameter $\psi(z)$ depends exclusively on the z-coordinate. In the absence of external magnetic fields, the one-dimensional Ginzburg-Landau equation is expressed as:

$$-\frac{\hbar^2}{2m^*}\frac{d^2\psi(z)}{dz^2} + \beta|\psi(z)|^2\psi(z) = -\alpha(T)\psi(z) \tag{135}$$

For temperatures below the critical point ($T < T_c$), we can normalize this expression by dividing all terms by $|\alpha(T)|$, which yields:

$$-\frac{\hbar^2}{2m^*|\alpha(T)|}\frac{d^2\psi(z)}{dz^2} + \frac{|\psi(z)|^2}{|\psi_\infty|^2}\psi(z) = \psi(z) \tag{136}$$

The coefficient of the second-order derivative term defines the Ginzburg-Landau coherence length, $\xi_{GL}(T)$, such that:

$$\xi_{GL}^2(T) = \frac{\hbar^2}{2m^*|\alpha(T)|} \tag{137}$$

This length scale **represents the distance over which the order parameter recovers from a localized inhomogeneity, such as a material boundary**. For a surface at $z = 0$ where the boundary conditions are $\psi(0) = 0$ and $\psi(\infty) = \psi_\infty$, the solution to the differential equation is:

$$\psi(z) = |\psi_\infty| \tanh\left(\frac{z}{\sqrt{2}\xi_{GL}(T)}\right) \tag{138}$$

This confirms that $\xi_{GL}(T)$ dictates the spatial evolution of the superconducting condensate as it approaches its bulk equilibrium value.

In the study of superconductors, the relationship between the coherence length $\xi_{GL}(T)$ and the magnetic penetration depth $\lambda_{\text{eff}}(T)$ is of fundamental importance. While both lengths diverge following a $(T_c - T)^{-1/2}$ dependence as the transition temperature is approached, their ratio remains a temperature-independent material constant. This dimensionless Ginzburg-Landau parameter, κ, is defined as:

$$\kappa = \frac{\lambda_{\text{eff}}(T)}{\xi_{GL}(T)} = \frac{m^* c}{|e^*|\hbar}\sqrt{\frac{\beta}{2\pi}} \tag{139}$$

The magnitude of κ serves as the **primary criterion for classifying superconductors as either Type-I or Type-II**. At an interface between normal and superconducting phases, the coherence length defines the region of partial condensation energy, whereas the penetration depth defines the region of partial magnetic field exclusion. The Ginzburg-Landau framework has proven remarkably effective in modeling diverse phenomena, ranging from boundary formation and thin film behavior to the effects of impurities on the superconducting state.

7.3 Magnetic flux quantization.

A fundamental manifestation of macroscopic quantum behavior in superconductors is the quantization of magnetic flux. When considering a multiply connected superconductor, such as the ring geometry depicted in Figure 14, the magnetic flux Φ threading the interior cannot take arbitrary values. Instead, it is constrained to integer multiples of the flux quantum $\Phi_0 = \frac{hc}{2e}$.

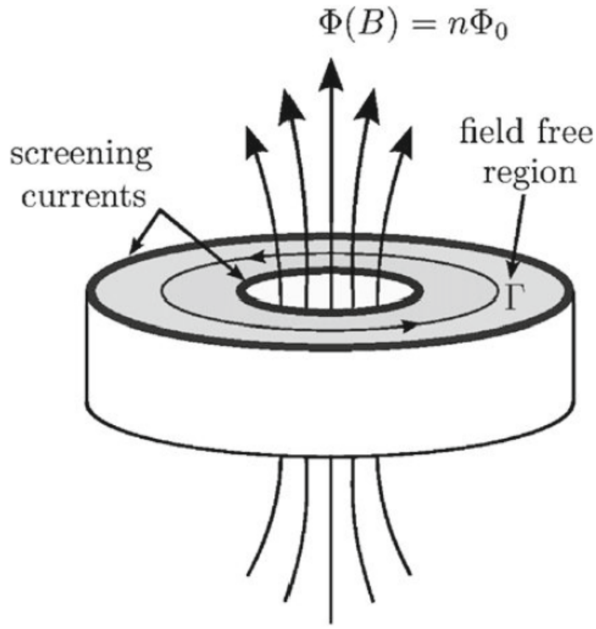


Figure 14: Quantization of the magnetic flux through a superconducting ring. The loop Γ is chosen deep within the bulk of the material where currents and fields vanish.

The origin of this quantization can be rigorously derived from the Ginzburg-Landau expression for the supercurrent density:

$$\vec{J}_s(\vec{r}) = \frac{e^*}{m^*}|\psi(\vec{r})|^2 \left(\hbar\vec{\nabla}\theta(\vec{r}) - \frac{e^*}{c}\vec{A}(\vec{r})\right) \tag{140}$$

where $\theta(\vec{r})$ represents the local phase of the macroscopic order parameter. Consider a closed loop Γ situated entirely within the bulk of the superconductor, **sufficiently far from the surfaces to ensure that both the magnetic field and the screening currents have decayed to zero due to the Meissner effect**. Within this region, $\vec{J}_s = 0$, which implies:

$$\vec{\nabla}\theta(\vec{r}) = \frac{2e}{\hbar c}\vec{A}(\vec{r}) \quad \text{for } \vec{r} \in \Gamma \tag{141}$$

In this derivation, we have substituted $e^* = -2e$ for the Cooper pair charge. By performing a line integral of both members around the closed circuit Γ, we obtain:

$$\oint_\Gamma \vec{\nabla}\theta(\vec{r}) \cdot d\vec{l} = \frac{2e}{\hbar c}\oint_\Gamma \vec{A}(\vec{r}) \cdot d\vec{l} \tag{142}$$

This leads to the relationship between the phase variation $\Delta\theta$ and the magnetic flux

Φ_Γ:

$$\Delta\theta = \frac{2e}{\hbar c}\Phi_\Gamma(\vec{B}) \tag{143}$$

where $\Phi_\Gamma(\vec{B})$ is the flux of the magnetic induction through the surface bounded by Γ. Because the order parameter $\psi(\vec{r})$ must be a single-valued function at any point in space, the total change in phase after completing the circuit must be an integer multiple of 2π ($\Delta\theta = 2\pi n$). Substituting this requirement into the previous expression yields the flux quantization condition:

$$\Phi_\Gamma(\vec{B}) = n\frac{hc}{2e} = n\Phi_0 \tag{144}$$

Here, n is an integer and Φ_0 represents the elementary fluxoid quantum. In a simply connected superconductor, the winding number n is necessarily zero to ensure regularity at the center. However, in multiply connected geometries or in materials containing non-superconducting inclusions, n can be any non-zero integer, reflecting the **topological nature of the superconducting state**.

7.4 Type I and type II superconductors.

The Ginzburg-Landau theory provides a rigorous classification of superconductors into two categories: Type-I and Type-II. As established previously, Type-I superconductors revert abruptly to the normal state when an external magnetic field exceeds the thermodynamic critical field $H_c(T)$. In contrast, Type-II superconductors exhibit a more complex behavior; they transition into a mixed state, or Shubnikov phase, for applied fields between the lower critical field $H_{c1}(T)$ and the upper critical field $H_{c2}(T)$. In this regime, the magnetic flux partially penetrates the material until the field reaches $H_{c2}(T)$, at which point the material becomes entirely normal.

To determine the upper critical field $H_{c2}(T)$, we examine the Ginzburg-Landau equation (Eq. 123) near the second-order phase transition. In this limit, the order parameter density $|\psi(\vec{r})|^2$ approaches zero, allowing for the linearization of the equation:

$$\frac{1}{2m^*}\left(\vec{p} - \frac{e^*}{c}\vec{A}\right)^2 \psi(\vec{r}) = -\alpha(T)\psi(\vec{r}) \tag{145}$$

This operator is mathematically equivalent to the Hamiltonian of a free particle with mass m^* and charge e^* in a uniform magnetic field H. The onset of superconductivity corresponds to the lowest eigenvalue of this system, which is the zero-point energy of the cyclotron motion $E_0 = \frac{1}{2}\hbar\omega_c = \frac{1}{2}\hbar\frac{|e^*|H}{m^*c}$. Equating this energy to $|\alpha(T)|$ allows us to define $H_{c2}(T)$ as:

$$\frac{1}{2}\hbar\frac{|e^*|H_{c2}(T)}{m^*c} = |\alpha(T)| \Rightarrow H_{c2}(T) = \frac{2m^*c}{|e^*|\hbar}|\alpha(T)| \tag{146}$$

By comparing this with the thermodynamic critical field $H_c(T)$, we obtain the following relationship involving the Ginzburg-Landau parameter κ:

$$\frac{H_{c2}(T)}{H_c(T)} = \sqrt{2}\kappa = \sqrt{2}\frac{\lambda_{\text{eff}}(T)}{\xi_{\text{GL}}(T)} \tag{147}$$

The critical threshold for Type-II behavior occurs at $\kappa = \frac{1}{\sqrt{2}}$. If $\kappa < \frac{1}{\sqrt{2}}$, the material is a Type-I superconductor, a category that includes most pure elemental metals. If $\kappa > \frac{1}{\sqrt{2}}$, the material is Type-II, which is typical for alloys and high-T_c cuprates (where κ can reach values near 100) due to the reduction of the mean free path.

In the mixed state, magnetic flux penetrates the superconductor in the form of quantized vortex tubes, each carrying a single flux quantum $\Phi_0 = \frac{hc}{2e}$. Each vortex consists of a normal core with a radius approximately equal to the coherence length ξ_{GL}, where the order parameter vanishes, surrounded by circulating supercurrents that extend to a distance of λ_{eff}. An estimate for the lower critical field $H_{c1}(T)$ can be obtained by assuming a single flux quantum is contained within a cylinder of radius λ_{eff}:

$$\pi\lambda_{\text{eff}}^2(T)H_{c1}(T) \sim \Phi_0 = \frac{hc}{2e} \Rightarrow H_{c1}(T) \sim \frac{hc}{|e^*|\pi\lambda_{\text{eff}}^2(T)} = 2\sqrt{2}\frac{H_c(T)}{\kappa} \tag{148}$$

Equations 146 and 148 demonstrate that the thermodynamic critical field $H_c(T)$ serves approximately as the geometric mean of the lower and upper critical fields.

8. Tunneling effects.

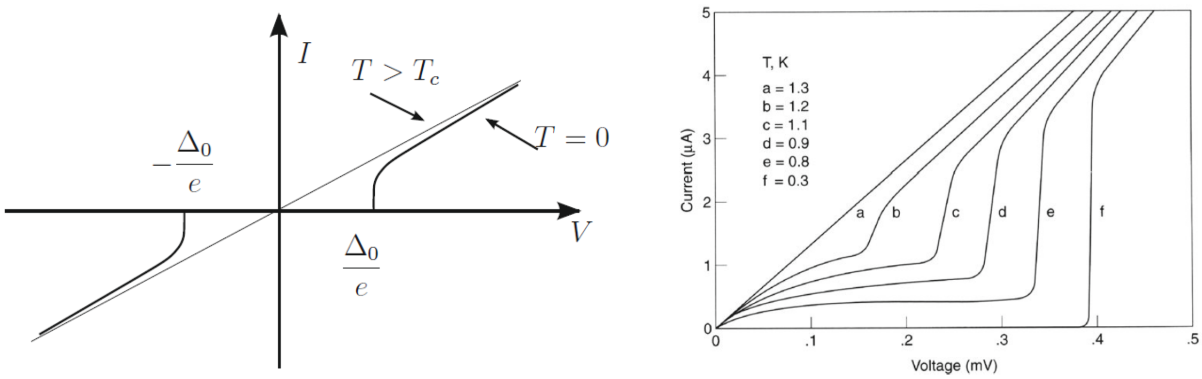
8.1 Electron tunneling into superconductors.

The phenomenon of electron tunneling into superconductors represents a cornerstone of the BCS theory, providing **the most direct experimental evidence for the existence of a finite energy gap in the electronic density of states**. Initially investigated by I. Giaever, these experiments involve junctions where metals are separated by a thin insulating barrier, typically in the range of 10 to 50 Å. In a junction composed of two normal metals, the tunneling current exhibits an ohmic dependence on the applied voltage V for low fields. Under such quasi-equilibrium conditions, the Fermi levels of the respective metals are shifted by the potential energy eV. Given that the density of states and the tunneling transition probabilities remain essentially constant within the millielectronvolt energy range surrounding E_F, the current flow remains proportional to the bias. However, when examining a normal metal-insulator-superconductor (NIS) junction at zero temperature, the physical behavior is fundamentally altered by the superconducting energy gap Δ_0. In the normal metal, states are occupied up to E_F with no gap present, whereas in the superconducting state, quasiparticle excitations are separated from the Fermi level by at least Δ_0. Consequently, no single-particle states are available for electron transfer until the applied bias potential satisfies the condition:

$$V > \frac{\Delta_0}{e} \tag{149}$$

As the voltage exceeds this threshold, the differential conductance $G = \frac{dI}{dV}$ directly reflects the quasiparticle density of states in the superconductor, eventually recovering ohmic behavior at high bias where $V \gg \frac{\Delta_0}{e}$. These characteristics are depicted in the schematic and experimental comparisons provided in Fig. 15.

In the case of a superconductor-insulator-superconductor (SIS) junction involving two identical superconductors with a gap parameter Δ_0, the threshold for quasiparticle tunneling at $T = 0$ is shifted to $V > \frac{2\Delta_0}{e}$. For junctions composed of two distinct superconducting materials, this threshold is generalized to the sum of their respective gaps, given by $\frac{\Delta_1+\Delta_2}{e}$. At this specific threshold voltage, the current exhibits a sharp discontinuous jump, a direct consequence of the singularities inherent in the quasiparticle density of states. At finite temperatures $0 < T < T_c$, thermally excited quasiparticles lead to a smearing of the $I - V$ curves, allowing for a non-zero current below the formal threshold, though the strongly non-linear features remain dominant. The recovery of the linear ohmic regime is again observed at sufficiently high bias voltages.



(a) NIS junction characteristics.

(b) SIS junction I-V curves.

Figure 15: Electronic tunneling characteristics in superconducting junctions: (a) schematic representation of the NIS junction behavior at $T = 0$ and $T > T_c$, and (b) typical quasiparticle tunneling current in an SIS junction showing the sharp break at $eV = 2\Delta(T)$.

8.2 Cooper pair tunneling and Josephson effects.

Beyond the quasiparticle tunneling mechanisms previously discussed, superconductor-insulator-superconductor (SIS) junctions with extremely thin insulating layers (typically in the range of 10 to 15 Å) exhibit a macroscopic quantum phenomenon known as Cooper pair tunneling. In such configurations, the electron pair correlations extend through the barrier, allowing for a sufficiently strong coupling to maintain a definite phase relationship between the superconducting condensates on opposite sides of the insulator. **As predicted by B. D. Josephson, this coupling enables the dissipative-free transport of paired electrons across the barrier**. The resulting supercurrent is highly sensitive to external electromagnetic fields and constitutes the foundation of the Josephson effects. In the dc Josephson effect, a direct supercurrent of pairs can flow across the junction with zero voltage drop ($V = 0$), provided the current does not exceed a characteristic maximum value I_J. The insulating barrier, while limiting the magnitude of this critical supercurrent, introduces no electrical resistance to the flow.

The current-voltage ($I - V$) characteristics of such a junction at $T = 0$ are summarized in Fig. 16, illustrating the distinct regimes of transport. For a bias voltage $V = 0$, the junction supports the dc Josephson current up to I_J. If a constant finite voltage V is established across the junction, the system enters the ac Josephson effect regime. In this state, an alternating supercurrent flows with a characteristic Josephson frequency defined by the relation:

$$\omega_J = \frac{2eV}{\hbar} \tag{150}$$

The instantaneous current is then given by $I_J \sin(\omega_J t + \phi_0)$, where ϕ_0 is an initial phase constant. It is observed that for voltages in the range $0 < V < \frac{2\Delta_0}{e}$, no direct supercurrent is detected due to the oscillatory nature of the ac term, and quasiparticle tunneling remains energetically suppressed. Only when the bias exceeds the gap threshold $V > \frac{2\Delta_0}{e}$ does the quasiparticle tunneling current become the dominant transport mechanism, eventually approaching the linear ohmic regime at high potentials.

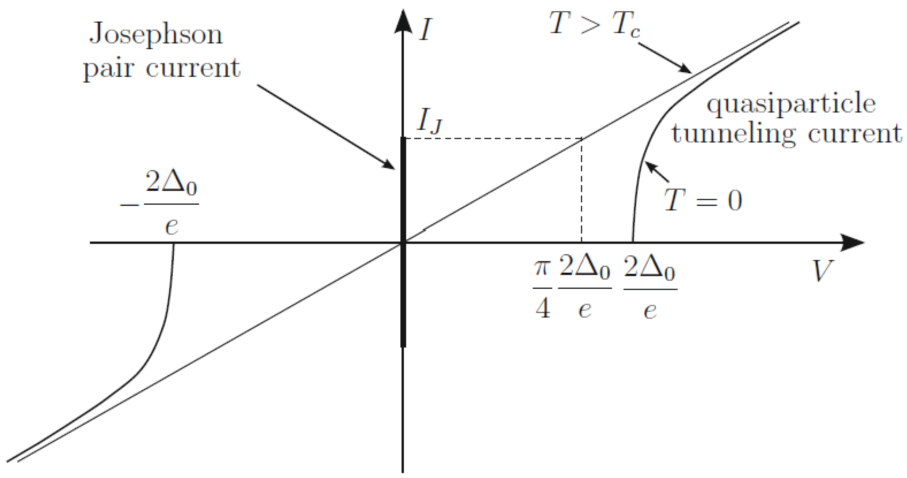


Figure 16: Schematic representation of the I-V characteristics for an SIS junction at $T = 0$ demonstrating the Josephson effect. The vertical line at $V = 0$ indicates the dc supercurrent up to I_J, while the region $V > \frac{2\Delta_0}{e}$ shows the onset of quasiparticle tunneling.

Weak links and Josephson effects.

The phenomenon of weak links and the associated Josephson effects can be analyzed by considering the macroscopic quantum behavior of two superconductors separated by a thin insulating barrier of width b, as illustrated in Fig. 17. In the limit where the barrier is infinitely thick, each superconductor is described by a rigid order parameter, $\psi_1 = |\psi_1|e^{i\theta_1}$ and $\psi_2 = |\psi_2|e^{i\theta_2}$, where the magnitudes and phases θ_i are uniform throughout their respective volumes. For a barrier of finite, sufficiently small thickness, the superconducting order parameters ψ_1 and ψ_2 penetrate the insulating region, decaying exponentially. The spatial variation of the order parameter $\psi(z)$ within the barrier ($0 < z < b$) can be approximated by a linear combination of these decaying evanescent waves:

$$\psi(z) = \psi_1 e^{-\beta z} + \psi_2 e^{\beta(z-b)} \tag{151}$$

where the parameter β characterizes the damping strength within the insulating medium.

To determine the resulting supercurrent density $J_S(z)$, we substitute the expression for $\psi(z)$ into the Ginzburg-Landau current equation, **assuming the limit of negligible magnetic fields**. The calculation proceeds as follows:

$$\begin{aligned} J_S(z) &= -\frac{i\hbar e^*}{2m^*}\left(\psi^*(z)\frac{d}{dz}\psi(z) - \psi(z)\frac{d}{dz}\psi^*(z)\right) = -\frac{i\hbar e^*}{m^*}\beta e^{-\beta b}\left(\psi_1^*\psi_2 - \psi_1\psi_2^*\right) = \\ &= \frac{2\hbar e^*}{m^*}\beta e^{-\beta b}|\psi_1||\psi_2|\sin\left(\theta_2 - \theta_1\right) \end{aligned} \tag{152}$$

This derivation highlights that the supercurrent is a periodic function of the phase difference $\delta = \theta_2 - \theta_1$ across the junction. The term multiplying the sine function represents the critical current density J_c, which depends exponentially on the barrier width b and the damping constant β. This relationship confirms that the Josephson current is a direct consequence of the phase coherence maintained across the weak link, with the damping in the insulator dictating the maximum current the junction can support without dissipation.

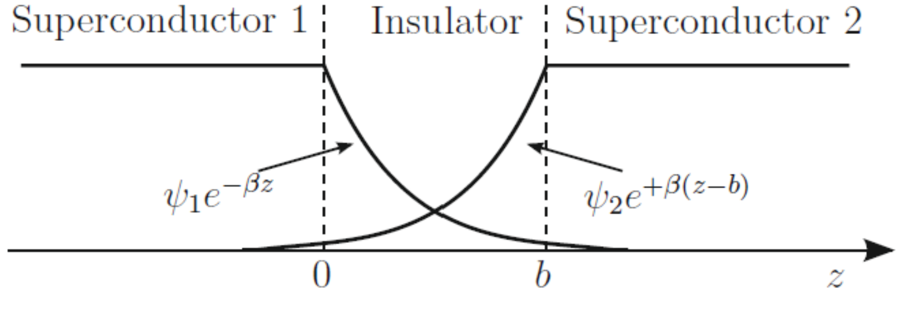


Figure 17: Schematic representation of a superconductor-insulator-superconductor junction. The diagram illustrates the exponential decay of the order parameter wavefunctions $\psi_1 e^{-\beta z}$ and $\psi_2 e^{\beta(z-b)}$ within the insulating barrier of width b.

The fundamental relationship governing the Josephson supercurrent across an insulating barrier is established by the phase difference $\gamma = \theta_2 - \theta_1$ of the order parameters in the adjacent superconductors. This current is expressed as:

$$I = I_J \sin \gamma \tag{153}$$

where I_J represents the critical current, a parameter contingent upon the specific geometrical and physical characteristics of the junction. While this relation can be motivated through heuristic arguments, a rigorous treatment within the microscopic BCS framework not only confirms this sinusoidal dependence but also provides the magnitude of I_J. For an ideal SIS tunnel junction composed of identical materials, I_J is equivalent to the current that would flow through the junction in its normal state when biased with a voltage equal to $\frac{\pi}{4} \cdot \frac{2\Delta_0}{e}$. This analysis assumes a negligible magnetic flux through the junction; in more general configurations, the dependence of I_J on the external magnetic field manifests as a characteristic diffraction pattern that must be accounted for.

When a potential difference V is established across the junction, the relative energy levels of the Cooper pairs on opposite sides are shifted by $2eV$. Consistent with the quantum mechanical evolution of a wave function, the rate of change of the relative phase γ is proportional to this energy difference, leading to the second Josephson relation:

$$\frac{d\gamma}{dt} = \frac{2eV}{\hbar} \tag{154}$$

These two equations collectively describe the dc and ac Josephson effects. Under a zero-bias condition ($V = 0$), the phase γ remains constant, allowing a stationary supercurrent with any intensity in the range $-I_J \leq I \leq I_J$ to flow without dissipation. This manifests as the characteristic zero-resistance spike observed in the $I - V$ profile. In a practical dc Josephson circuit, as shown in Fig. 18, a junction is placed in series with a resistance R and a voltage source V_0. If the condition $I = \frac{V_0}{R} < I_J$ is met, the system enters a stationary regime where the supercurrent flows with zero potential drop across the insulating layer.

Conversely, if a constant finite potential V is maintained across the junction within the sub-gap regime $0 < V < \frac{2\Delta_0}{e}$, the integration of the phase evolution equation yields a time-dependent phase $\gamma(t) = \frac{2eV}{\hbar} t + \gamma_0$. Substituting this result into the constitutive current-phase relation:

$$I(t) = I_J \sin \left(\frac{2eV}{\hbar} t + \gamma_0 \right) \tag{155}$$

reveals an alternating current that oscillates at the Josephson frequency without a net dc component, highlighting the fundamental coupling between the macroscopic phase and electromagnetic potentials. This transition from a stationary to an oscillatory state is practically utilized in the dc Josephson circuit, as illustrated in Fig. 18, where a direct supercurrent $I \leq I_J$ flows without dissipation provided the bias conditions are met.

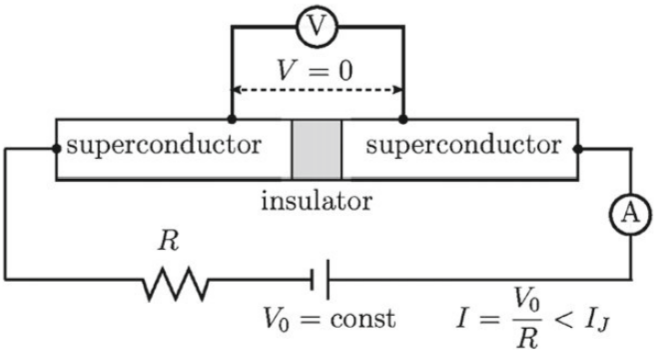


Figure 18: Schematic illustration of the dc Josephson circuit. A direct supercurrent flows through the insulating barrier, sustained by the phase coherence of the Cooper pair condensates.

This linear temporal evolution of the phase difference establishes the Josephson constant $K_J = \frac{2e}{h}$, a parameter of paramount importance in metrology with a value of approximately 483.5979 MHz/μV. **Experimental verification of the ac Josephson effect is typically conducted by superposing a radiofrequency or microwave voltage $V_r \cos \omega_r t$ onto the constant dc bias V.** Under this combined potential $V_{\text{tot}} = V + V_r \cos \omega_r t$, the dynamic equation for the phase is generalized to:

$$\frac{d\gamma}{dt} = \frac{2eV}{\hbar} + \frac{2eV_r}{\hbar} \cos \omega_r t \tag{156}$$

Upon integration, the time-dependent phase is given by:

$$\gamma(t) = \frac{2eV}{\hbar} t + \frac{2eV_r}{\hbar \omega_r} \sin \omega_r t + \gamma_0 \tag{157}$$

resulting in a frequency-modulated Josephson current

$$I(t) = I_J \sin \left(\frac{2eV}{\hbar} t + \gamma_0 + \frac{2eV_r}{\hbar \omega_r} \sin \omega_r t \right) \tag{158}$$

To analyze the spectral components of this current, we employ the Jacobi-Anger expansion for Bessel functions of the first kind J_n, utilizing the identity $\sin(b + a \sin x) = \sum_{n=-\infty}^{+\infty} J_n(a) \sin(b + nx)$. The current expression is thus transformed into:

$$I(t) = I_J \sum_{n=-\infty}^{+\infty} J_n \left(\frac{2eV_r}{\hbar \omega_r} \right) \sin \left(\frac{2eV}{\hbar} t + \gamma_0 + n\omega_r t \right) \tag{159}$$

A stationary, zero-frequency supercurrent component emerges whenever the resonance condition $2eV = n\hbar\omega_r$ is satisfied for any integer n. These resonances manifest as discrete, vertical current spikes in the $I - V$ characteristic, known as **Shapiro steps**. The magnitude of these steps is modulated by the Bessel functions as $I_J J_n \left(\frac{2eV_r}{\hbar \omega_r} \right)$. Specifically, the critical current at zero voltage is modified to $I_c = I_J J_0\left(\frac{2eV_r}{\hbar\omega_r}\right)$. Due to their extreme precision, Shapiro steps are utilized for the exact determination of the e/h ratio and have significant implications for the development of high-speed, lossless cryogenic switching elements in quantum computing.

Superconducting Quantum Interference Devices (SQUIDs).

The principles of macroscopic quantum interference are most elegantly demonstrated in Superconducting Quantum Interference Devices (SQUIDs). Consider a **superconducting loop containing two Josephson junctions**, P and Q, arranged in parallel as shown in Fig. 19. Assuming identical junctions with critical current I_J and neglecting the magnetic flux through the junctions themselves, we define γ_P and γ_Q as the phase differences across each weak link. Specifically, $\gamma_P = \theta(P_2) - \theta(P_1)$ and $\gamma_Q = \theta(Q_2) - \theta(Q_1)$, where θ represents the phase of the macroscopic superconducting wavefunction. The total current I traversing the device is the sum of the individual supercurrents:

$$I = I_J\left(\sin\gamma_P + \sin\gamma_Q\right) = 2I_J\sin\left(\frac{\gamma_P+\gamma_Q}{2}\right)\cos\left(\frac{\gamma_P-\gamma_Q}{2}\right) \tag{160}$$

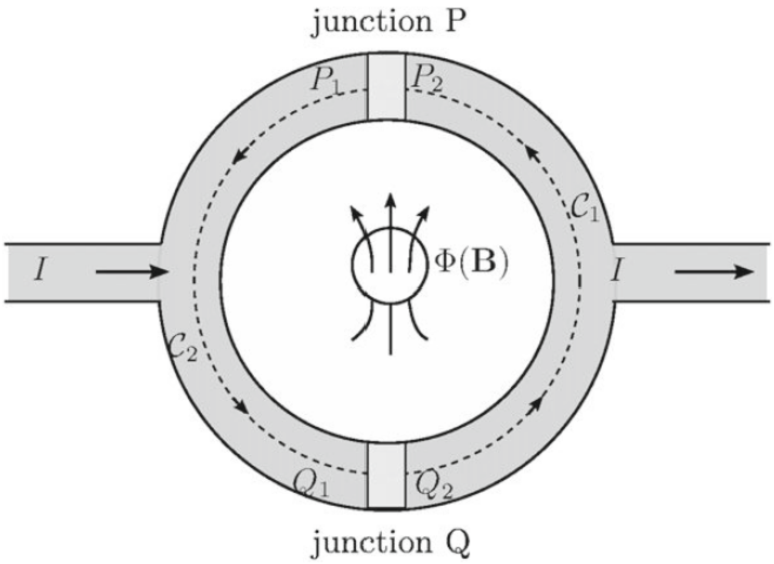


Figure 19: Schematic of a dc SQUID. The device consists of a superconducting loop with two parallel Josephson junctions, enabling the detection of an enclosed magnetic flux $\Phi(B)$ through quantum interference.

In the interior of the superconducting segments, where the Meissner state ensures $J_S = 0$, the gradient of the phase is coupled to the vector potential $\vec{A}(\vec{r})$ according to the relation $\vec{\nabla}\theta(\vec{r}) = -\frac{2e}{\hbar c}\vec{A}(\vec{r})$. By performing line integrals along the paths C_1 and C_2 connecting the junctions, the phase differences are linked to the magnetic flux $\Phi(B)$

enclosed by the loop. The sum of these integrals yields:

$$\theta(P_2) - \theta(Q_2) + \theta(Q_1) - \theta(P_1) = -\frac{2e}{\hbar c} \oint \vec{A}(\vec{r}) \cdot d\vec{l} \tag{161}$$

which implies that the relative phase shift is given by $\gamma_P - \gamma_Q = -2\pi \frac{\Phi(B)}{\Phi_0}$, where $\Phi_0 = \frac{hc}{2e}$ is the magnetic flux quantum. Substituting this into the current expression, we obtain:

$$I = 2I_J \cos\left(\frac{\pi\Phi(B)}{\Phi_0}\right) \sin\left(\frac{\gamma_P + \gamma_Q}{2}\right) \tag{162}$$

The maximum dissipationless current that the SQUID can support, $I_{\max}$, is thus a periodic function of the enclosed magnetic flux:

$$I_{\max}(\Phi) = 2I_J \left| \cos\left(\frac{\pi\Phi(B)}{\Phi_0}\right) \right| \tag{163}$$

This extreme sensitivity to flux changes, on the order of $\frac{\Phi_0}{2}$, allows SQUIDs to measure exceptionally weak magnetic fields. **Such precision is leveraged in biomagnetism for clinical applications like magneto-encephalography (MEG) and magneto-cardiography (MCG)**, where arrays of SQUIDs detect the minute magnetic signatures generated by physiological electrical activity.

Appendix. Phonon-induced electron-electron interaction.

The fundamental origin of conventional superconductivity lies in a subtle effective attractive interaction between electrons, a phenomenon generally attributed to the Fröhlich mechanism mediated by the lattice phonon field. While a rigorous quantitative characterization of such effective interactions presents a formidable many-body problem, the essential physics can be captured through qualitative models that highlight the emergence of attraction for electronic states in close proximity to the Fermi surface. Specifically, this indirect interaction becomes dominant for electrons situated within an energy shell of order $\hbar\omega_D$ centered at E_F, where ω_D denotes the Debye frequency of the ionic lattice.

To provide a plausible physical basis for this attractive potential, one may employ a well-established canonical transformation. This mathematical framework allows for the decoupling of the direct electron-phonon coupling into an effective retarded interaction between the electrons themselves. By transforming the Hamiltonian into a representation where the first-order phonon-mediated processes are expressed as second-order electronic scattering events, it becomes evident that the exchange of virtual phonons can overcome the screened Coulomb repulsion under specific energy-momentum conditions. This mechanism facilitates the formation of Cooper pairs, which constitute the superconducting condensate.

A canonical transformation.

To rigorously derive the effective interaction, we consider a Hamiltonian H and perform a canonical transformation of the form:

$$\begin{aligned}\tilde{H} = e^{-S} H e^{S} &= \left(1 - S + \frac{1}{2!}S^2 - \dots\right) H \left(1 + S + \frac{1}{2!}S^2 + \dots\right) = \\ &= H + [H, S] + \frac{1}{2!}[[H, S], S] + \frac{1}{3!}[[[H, S], S], S] + \dots\end{aligned} \tag{164}$$

where S is an arbitrary antihermitian operator ($S^\dagger = -S$). This requirement ensures that the transformation $U = e^S$ is unitary, thereby preserving the norm of the states and the eigenvalues of the system. We partition the Hamiltonian into an unperturbed part H_0 and a perturbation H_1, such that $H = H_0 + H_1$. The transformed Hamiltonian then expands as:

$$\tilde{H} = H_0 + H_1 + [H_0, S] + [H_1, S] + \frac{1}{2}[[H_0, S], S] + \dots \tag{165}$$

By exploiting the degree of freedom in S, we can eliminate the first-order term in the perturbation by requiring that $H_1 + [H_0, S] = 0$. Under this condition, the transformed Hamiltonian simplifies to:

$$\tilde{H} = H_0 + \frac{1}{2}[H_1, S] + \dots \tag{166}$$

In this representation, $\tilde{H} = H_0 + H_{\text{indirect}}$, where the indirect interaction term $H_{\text{indirect}} = \frac{1}{2}[H_1, S]$ accounts for the second-order corrections in the coupling parameter. In a basis where H_0 is diagonal, the matrix elements of S are determined by:

$$\langle n|S|m\rangle = \frac{\langle n|H_1|m\rangle}{E_m - E_n} \tag{167}$$

where we assume the diagonal elements of H_1 are zero or absorbed into H_0. The matrix elements of the indirect interaction between an initial state $|i\rangle$ and a final state $|f\rangle$ are then given by:

$$\begin{aligned}\langle f|H_{\text{indirect}}|i\rangle &= \frac{1}{2}\langle f|H_1 S - S H_1|i\rangle = \frac{1}{2}\sum_\alpha \left(\langle f|H_1|\alpha\rangle\langle\alpha|S|i\rangle - \langle f|S|\alpha\rangle\langle\alpha|H_1|i\rangle\right) = \\ &= \frac{1}{2}\sum_\alpha \langle f|H_1|\alpha\rangle\langle\alpha|H_1|i\rangle \left(\frac{1}{E_i - E_\alpha} + \frac{1}{E_f - E_\alpha}\right)\end{aligned} \tag{168}$$

This expression demonstrates that **the indirect interaction arises from transitions through intermediate states $|\alpha\rangle$**. Notably, the diagonal elements $\langle i|H_{\text{indirect}}|i\rangle$ recover the standard second-order energy correction from perturbation theory, while the off-diagonal elements describe the renormalization of the interaction through virtual processes.

Indirect electron-electron interaction.

The microscopic mechanism for the indirect electron-electron interaction, mediated by the exchange of virtual phonons, is conceptually illustrated through the scattering processes shown in Fig. 20. In the primary process, an electron with wavevector $\vec{k}_1$ emits a phonon of wavevector $\vec{q}$, scattering into the state $\vec{k}_1' = \vec{k}_1 - \vec{q}$. This phonon is subsequently absorbed by a second electron with wavevector $\vec{k}_2$, which then transitions into $\vec{k}_2' = \vec{k}_2 + \vec{q}$. A symmetric process occurs when the second electron emits a phonon $-\vec{q}$ that is later absorbed by the first. To quantify this interaction, we define the initial, intermediate, and final states of the electron-phonon system as:

$$\begin{aligned} |i\rangle &= |\vec{k}_1,\, \vec{k}_2;\, \text{Vac}\rangle, \quad E_i = E(\vec{k}_1) + E(\vec{k}_2) \\ |\alpha\rangle &= |\vec{k}_1',\, \vec{k}_2;\, 1_{\vec{q}}\rangle, \quad E_\alpha = E(\vec{k}_1') + E(\vec{k}_2) + \hbar\omega_{\vec{q}} \\ |f\rangle &= |\vec{k}_1',\, \vec{k}_2';\, \text{Vac}\rangle, \quad E_f = E(\vec{k}_1') + E(\vec{k}_2') \end{aligned} \tag{169}$$

where Vac and $1_{\vec{q}}$ denote the phonon vacuum and single-phonon occupancy states, respectively. Applying the canonical transformation to these states, the indirect interaction matrix element is derived from the transition amplitudes $M_{\vec{q}}$ as:

$$\langle f|H_{\text{indirect}}|i\rangle = \frac{1}{2}|M_{\vec{q}}|^2 \left(\frac{1}{E(\vec{k}_1) - E(\vec{k}_1') - \hbar\omega_{\vec{q}}} + \frac{1}{E(\vec{k}_2') - E(\vec{k}_2) - \hbar\omega_{\vec{q}}} \right) \tag{170}$$

By summing the contributions from both possible time-orderings of the exchange, the total effective interaction potential is obtained:

$$\langle f|H_{\text{indirect}}|i\rangle = |M_{\vec{q}}|^2 \left(\frac{\hbar\omega_{\vec{q}}}{(E(\vec{k}_1) - E(\vec{k}_1'))^2 - \hbar^2\omega_{\vec{q}}^2} + \frac{\hbar\omega_{\vec{q}}}{(E(\vec{k}_2') - E(\vec{k}_2))^2 - \hbar^2\omega_{\vec{q}}^2} \right) \tag{171}$$

For the specific case of interest in BCS theory, namely the scattering of a Cooper pair from states $(\vec{k}, -\vec{k})$ to $(\vec{k}', -\vec{k}')$, the effective potential $V_{\vec{k}\vec{k}'}$ simplifies to:

$$V_{\vec{k}\vec{k}'} = |M_{\vec{q}}|^2 \frac{2\hbar\omega_{\vec{q}}}{(E(\vec{k}) - E(\vec{k}'))^2 - \hbar^2\omega_{\vec{q}}^2} \tag{172}$$

This potential is clearly attractive when the electronic energy difference $|E(\vec{k}) - E(\vec{k}')|$ is less than the phonon energy $\hbar\omega_{\vec{q}}$. In the vicinity of the Fermi surface, within an energy shell of the order of the Debye energy $\hbar\omega_D$, this phonon-induced attraction can effectively overcome the screened Coulomb repulsion, providing the binding glue necessary for the formation of the superconducting state.

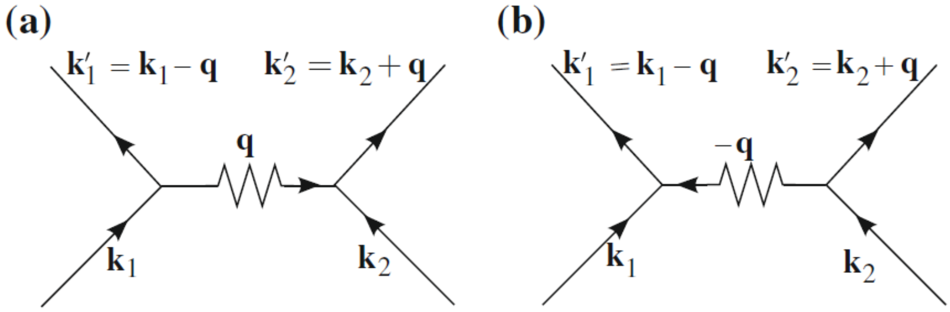


Figure 20: Diagrammatic representation of the indirect electron-electron interaction. (a) Electron $\vec{k}_1$ emits a phonon $\vec{q}$ which is absorbed by $\vec{k}_2$; (b) Electron $\vec{k}_2$ emits a phonon $-\vec{q}$ which is absorbed by $\vec{k}_1$.